

FORM THREE EXAMS SET 3



SEVERAL PAPERS PER SUBJECT!

FOR ANSWERS AND MORE SETS OF EXAMS, SMS OR WHATSAPP: 0723981225

NAME.....ADMISSION NUMBER.....

AGRICULTURE

TIME:2 HOURS

2020 FORM 3 TERM 1 OPENER EXAMS

AGRICULTURE

TERM 3

INSTRUCTIONS TO CANDIDATES.

- Write your name and admission number in the spaces provided.
- Answer all questions in the spaces provided.
- This question paper consists of three sections A, B and C.
- Answer all the questions in sections A,B and C.
- Answers should be written in the spaces provided.

For Examiner's use only

Question	Marking	Maximum score	Candidate's score

Candidates should check the question paper to ascertain that all the pages are printed as indicate and no questions are missing.

SECTION A(30MKS)

1. Give four reasons why Agriculture is a art?

2mks

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2. State two branches of livestock farming.

1mk

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3. State four human factors which influence agricultural production.

2mks

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4. Give two aspects if light which affect agricultural production.

2mks

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5. Give any two light breeds of poultry 1mk

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6. State four importance of primary cultivation.

2mks

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7. (a)Give two sources of underground waters.

2mks

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(b) State two conditions under which irrigation is necessary. 1mk

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8. (a) State three characteristics of crops used for green manure. 1½mks

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(b) Give three reasons as to why green manure is not commonly used. 1½mks

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9. State the three basic concepts of economics . 1½mk

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10. Give three function of vitamins in animals body 1½mks

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11. Differentiate between micro and macro nutrients. 2mks...

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12. State two ways under which fertilizer are classified. 1mk

13. State four advantages of using seeds as a planting materials. 4x½=2mks

14. Give four methods of breaking seed dormancy 2mks

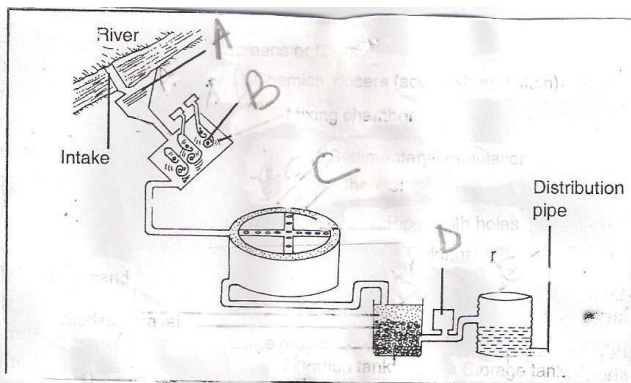
15. State four advantages of raising seedlings in a nursery. 2mks

16. Define the following terms as used in crop production. 2mks

Gapping.....
Rogueing.....
Staking.....
Thinning.....

SECTION B(20MKS)

17. The diagram below shows a chemical treatment of water .study the diagram and answer the questions that follow.



a. Name the stages labeled A,B ,C and D. 2mks

A.....

B.....
C.....
D.....

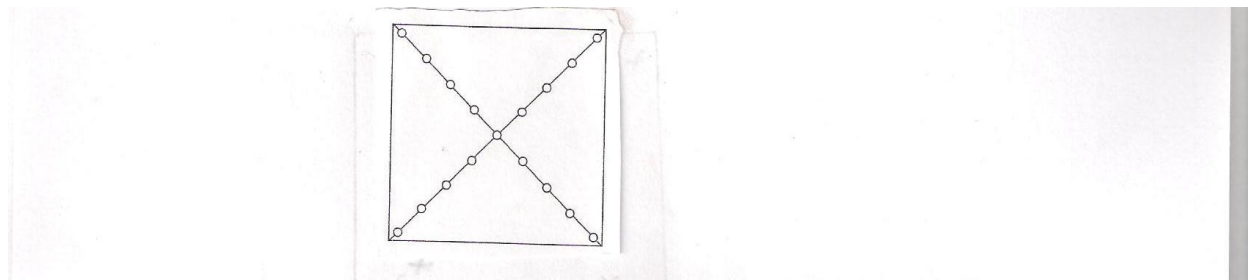
b. Name the chemical which are added in B. 1mks

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c. Give a reason as to why chemical added in C use in water treatment. 1mk

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18. The diagram below shows a method of soil sampling



a. Name the method of soil sampling shown above. 1mk

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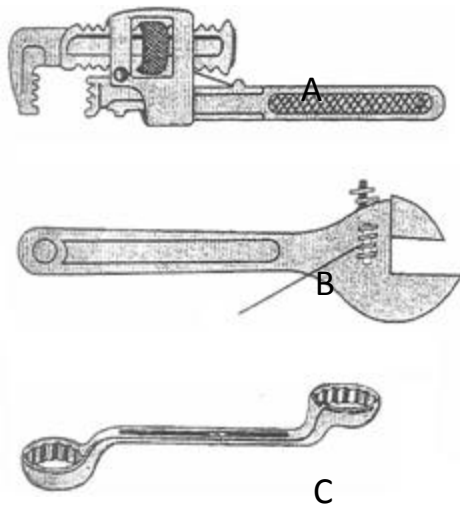
b. Besides the method shown above in the diagram, name the other method one can use to perform soil sampling. 1mk

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c. State the information that should be provided on the soil sampling package before it is taken to the laboratory. 3mks

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19. The diagram below show some workshop tools. Study the diagrams and answer the questions that follow.



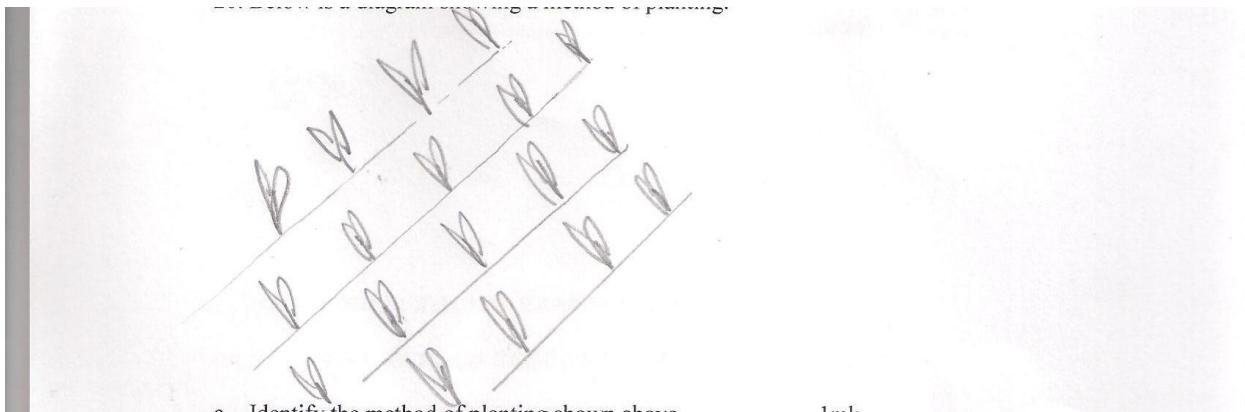
- (a) Identify tools labelled. (3 marks)
- A
B
C

(b) State the function of the tool labelled A.

(1 mark)

- (c) State the function of part X in the tool labelled B. 1mk

20. Below is a diagram showing a method of planting.



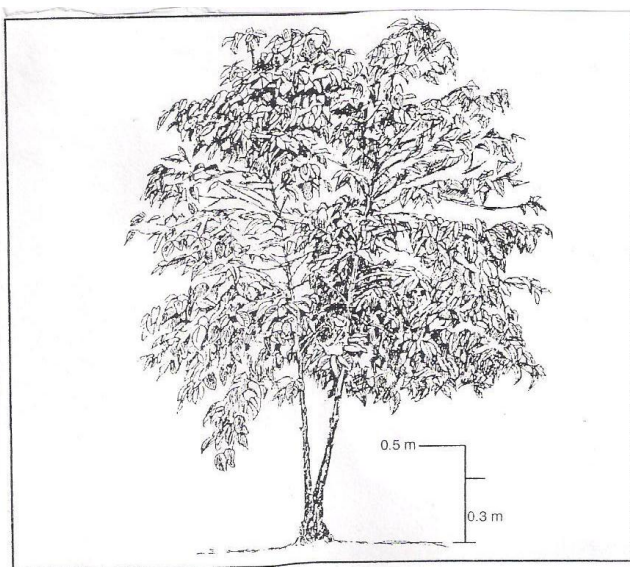
a. Identify the method of planting shown above.

1mk

b. State four advantages of using the method above.

2mks

21. The diagram below illustrate a method of pruning coffee .Study it carefully and answer the questions that follow.



a. Identify the method above.

½mk

b. Apart from the method named above, name the other method of pruning coffee.½mk

c. State two disadvantages of using the above method.

2mks

SECTION C(40 MKS)

ANSWER TWO QUESTIONS ONLY.

22. A)State the effect of wind in agricultural production.8mks

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d)Discuss 3 sources of water in an animals body 3mks

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23. (a)Explain various practices which are carried out in a nursery. 10mks

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b)Describe the factors influencing the quality of farm yard manure. 5mks

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c) Explain the importance of using the following materials when making a compost manure.

- i. Wood ash.....
- ii. Top soil.....
- iii. Well rotten manure.....
- iv. maize straws.....
- v. Tester.....

24. A) Describe Four predisposing factors of livestock diseases

4 mks.

b) outline the advantages of pruning in crop production.

4mks

c) Outline Four importance of keeping livestock health

2mks

NAME.....ADM NO.....

BIOLOGY

2020 FORM 3 TERM 1 OPENER EXAMS

TIME: 2HRS

INSTRUCTIONS

- I. Write your name and admission number in the space provided.
2. Answer all the questions in the space provided

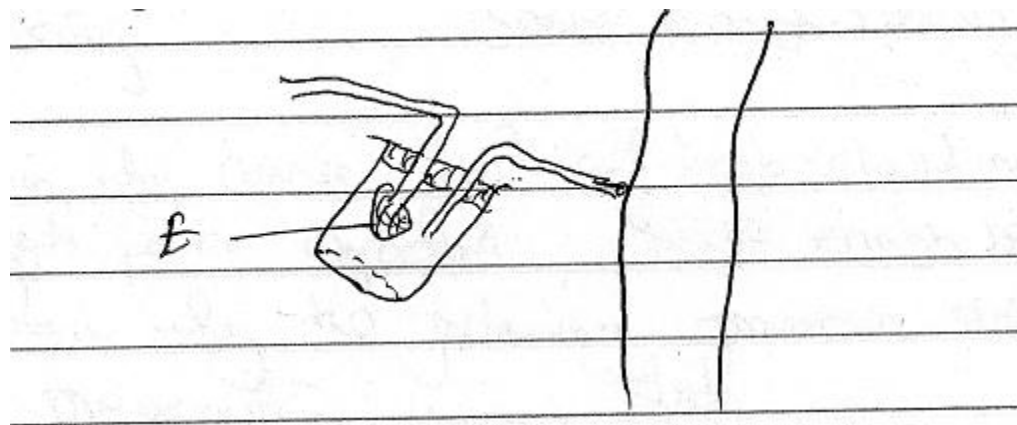
For examiner use only

Question No.	Total score	Student score
1-28	100	

1(a) What is Biology? (1mk)

(b) State and define any two main branches of biology. (2mks)

2 Below is an illustration of a piece of apparatus strategically positioned to trap some organism



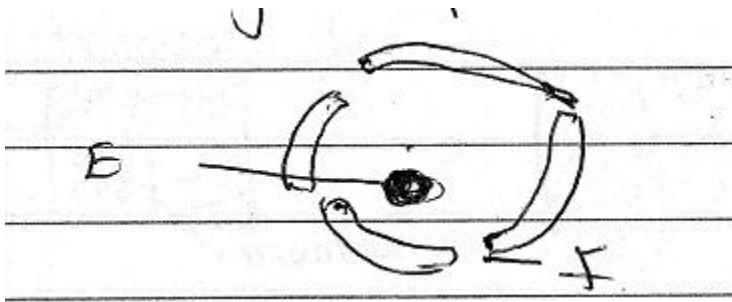
(a) Name the apparatus. (1mk)

- (b) State the function of the part labeled E. (1mk)
3. State the name given to the study of (2mks)
- (a) Micro-organism
- (b) Living things and their surrounding
- 4 State the organelles that (2mks)
- (i) provide site for protein synthesis
- (ii) involved in cell division through formation of spindle fibre
- 5(a) Give two precautions that should be observed when placing a microscope on a bench. (2mks)
- (b) To estimate the size of a cell A student observed 20 cells across a diameter of field of view of 5mm in diameter
Calculate the cell diameter . Show your working. (2mks)
- (c) Explain why course adjustment knob should not be used to lower the high power objective. (1mk)

6(a) Explain why the following processes are important during the preparation of a slide. (2mks)

- (i) making a very thin section
- (ii) placing a cover slip over the section

(b) The diagram represents a nucleus



(i) State the function of part labeled E (1mk)

(ii) Name part labeled F. (1mk)

7. List two differences between plant and animals. (2mks)

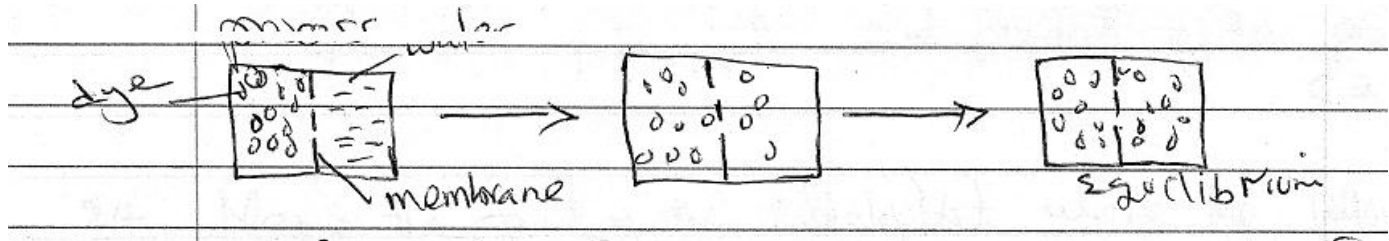
Plants	Animals

8. A student named cockroach as AMERICAN PERIPLANTA.

(a) Identify two mistakes made in naming the cockroach. (2mks)

(b) Write the name correctly. (1mk)

9 The set up below illustrate a certain physiological process.

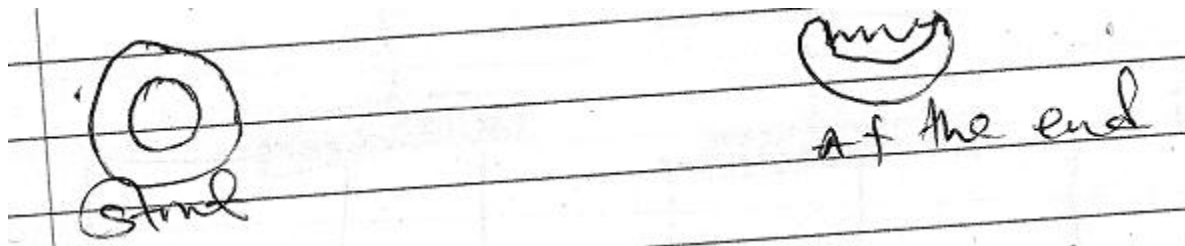


(i) Name the physiological process. (1mk)

(ii) Give two ways in which the movement of dye can be increased. (2mks)

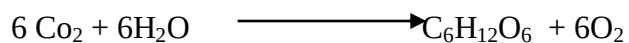
(b) List at least three factors that affect the process named above. (3mks)

10 The diagram below show red blood cell that was subjected to certain treatment.



- (a) Give the name given to cells appearance at the end. (1mk)
- (b) Name the solution in which the cell was placed in (1mk)

11 The reaction below represent a process that takes place in plants.



- (a) Name the process (1mk)
- (b) State two factors affecting the process named above. (2mks)
- (c) Explain the fate of the product of the process named above (2mks)
- 12 Name the part of the chloroplast where the following occur (2mks)
- (i) Carbon (IV) oxide fixation

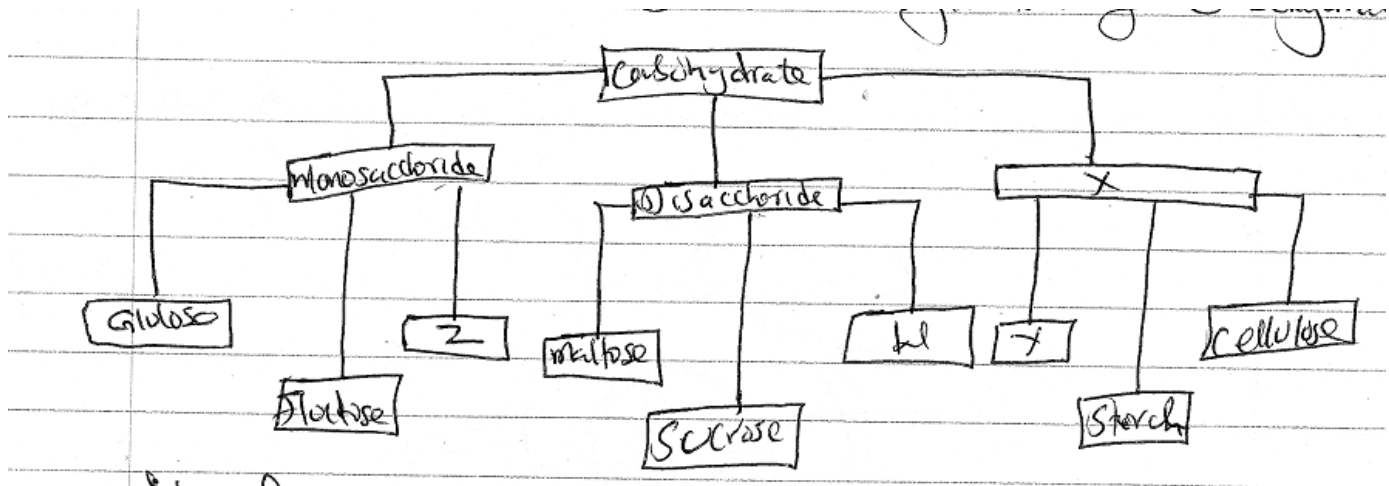
(ii) photolysis

13 Name three plant cells that contain chloroplasts

(3mks)

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14 The flow chart below shows classification of carbohydrates.



(a) Identify the form of carbohydrates represented by letters W,X,Y and Z. (4mks)

W

X

Y

Z

15(a) Differentiate between homodont and heterodont teeth

(2mks)

(b) Name two enzyme in the digestive system secreted in their inactive form.

(2mks)

(c) State the name of the juice secreted in the following region, (2mks)

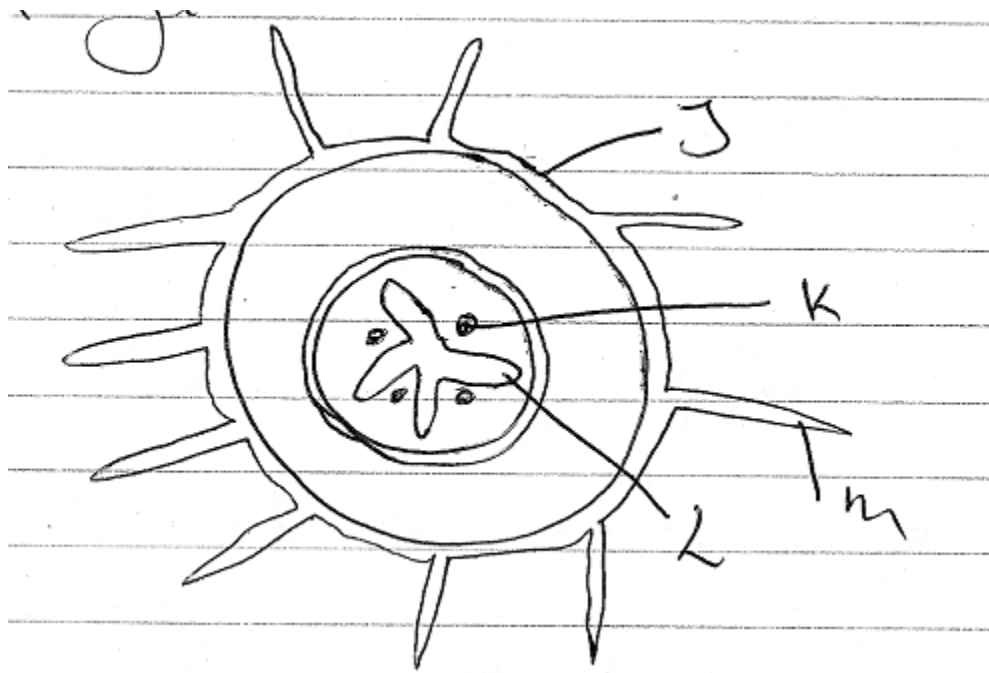
(i) stomach

(ii) small intestine

d) State two functions of saliva. (2mk)

(e) What is the importance of including roughages in the diet. (1mk)

16 The diagram below represent a transverse section of a plant organ



a) From which plant organ was the section obtained. (1mk)

b) Give two reasons for your answer in (a) above. (2mk)

c) Name the part labeled: (3mks)

J -

K -

L -

17(a) State why it is important for plant to lose water to the atmosphere. (2mks)

(b) Give the importance of mitochondria found in the companion cell of phloem. (1mk)

18(a) State two function of blood other than transport. (2mks)

b) State two forms in which carbon(IV) oxide is transported. (2mks)

19 State the role of the following during blood clotting process (2mks)

(a) Thromboplastin

(b) Calcium ions

20 Name two gaseous exchange structures in higher plants. (2mks)

21 What is a single circulatory system. (1mk)

22 The table provided shows the transportation of substances in the human body.

Substance	Transported by blood	
	From	To
Oxygen	M	Heart
N	Liver	Kidney
P	Intestine	All body

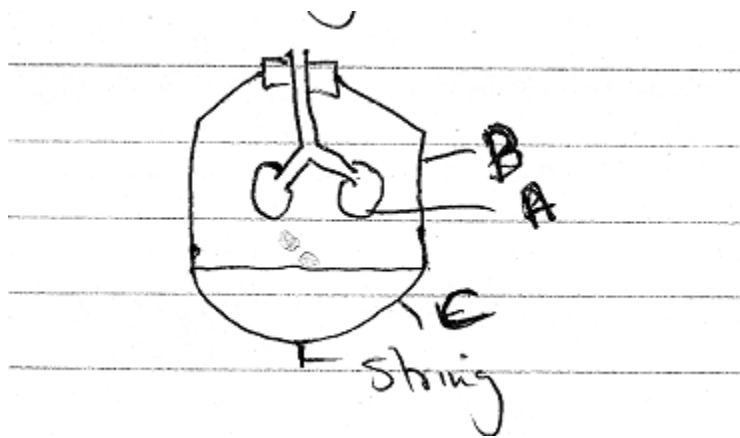
Name substance (3marks)

M

N

P

23 The diagram below represent a model used to demonstrate breathing in mammals.



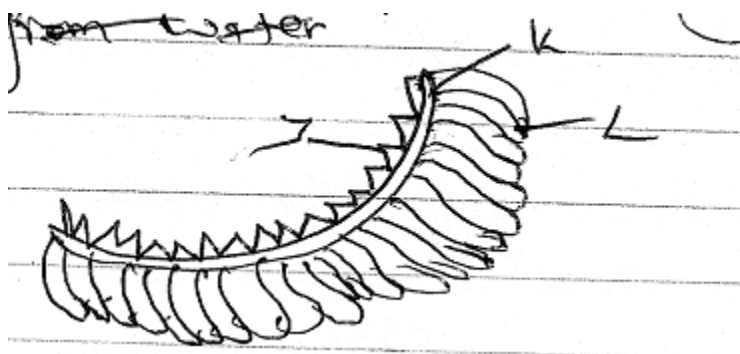
a) Name the mammalian structure represented by the part labeled A and B. (2mks)

A

B

24(a) Name three characteristics of a gaseous exchange surface. (3mks)

(b) The diagram below represent an organ in a bony fish.



(i) Name parts labeled (2mks)

K -

J -

(ii) Explain why bony fish dies shortly after being removed from the water. (2mks)

25(a) Define the term respiration. (1mk)

(b) Give three products of anaerobic respiration in plants. (3mks)

26(a) What is oxygen debt (1mk)

(b) What happens to lactic acid formed in muscle during excersice when there is sufficient oxygen. (2mks)

(c) Write an equation to summarize the process of aerobic respiration. (1mk)

27(a) State two ways in which plant manage their waste. (2mks)

(b) State one function of each of the following plant product. (2mks)

(i) Papain

(ii) Colchicine

NAME.....ADMISSION NUMBER.....

565/1

FORM THREE

BUSINESS STUDIES 565/1

TIME:2 HOURS

2020 FORM 3 TERM 1 OPENER EXAMS

Instructions to candidates.

- Write your name and admission number in the spaces provided.
- Answer all questions in the spaces provided.
- Answer **all** questions in Section A and **three** in Section B
- Answers must be written in English.
- Avoid one word answers

Total	
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Section A: Attempt all questions in this section

1. Give any four functions of an office. (4 Mks)

- a.
- b.
- c.
- d.

2. Distinguish between the following: (4Mks)

a. Producer goods and Consumer Goods

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b. Intermediate goods and Finished goods

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3. The following descriptions relates to some documents used in Home trade. Identify the document that befits the description(4mks)

Purpose of the document	Name of the document
To inform the buyer that goods have been dispatched	
To inform the buyer that the order has been received	
Evidence of delivery of goods	
Corrects an overcharge on an invoice	

4. In each of the following statement give the type of utility created. (4mks)

Statement

Type of utility

- a) Student buying a pen from a school canteen:
- b) A farmer taking mangoes to the market:.....
- c) Kamau banking money with Equity bank account:.....
- d) Making a table at carpentry shop:.....

5. Outline four fundamental features of a good transport system. (4 Mks)

- a.
- b.
- c.
- d.

6. Below are tasks performed by office machines and equipment. State the machine/equipment used to perform each task (4mks) (4mks)

Task	Machine/equipment
a) Sending and receiving verbal messages	
b) Cutting papers into required sizes	
c) Printing stamp impression on envelops	
d) Sending printed diagrams charts, maps e.t.c.	

7. State four consequences that may arise due to poor arrangement of goods in a warehouse. (4 Mks)

- a.
- b.
- c.
- d.

8. Highlight four Ways in which legal political environment may improve business performance. (4mks)

- a.
- b.
- c.
- d.

9. State four services offered by wholesalers to retailers (4 Mks)

- a.
- b.
- c.
- d.

10. State four functions of an entrepreneur. (4 Mks)

- a.
- b.
- c.
- d.

11. Outline four benefits of a bonded warehouse. (4 Mks)

- a.
- b.
- c.
- d.

12. Outline four difference between a public limited company and a partnership. (4 Mks)

	Public limited company	partnership
i.		
ii.		

iii.		
iv.		

a. ...

b.

13. State three measures a consumer may take in order to satisfy most of his wants. (3 Mks)

a.

b.

c.

d.

14. Highlight four factors to consider when choosing a method of promoting their product. (4 Mks)

a.

b.

c.

d.

15. State three circumstances when containers may be used in transport of goods. (3 Mks)

a.

- b.
- c.

16. Outline four contents of a business plan (4Mks)

- a.
- b.
- c.
- d.

17. State four reasons why a trader may prefer to advertise of radio over newspaper. (4 Mks)

- a.
- b.
- c.
- d.

18. Outline four circumstances when a partnership may be dissolved. 4 Mks)

- a.
- b.
- c.
- d.

Section B: Attempt any three questions (30 Marks)

19. A shop keeper ordered some items from a wholesaler. However, when the goods arrived, the shopkeeper found out that they were not the items ordered. Describe five reasons for this breakdown in communication between the two business people. (10 Mks)

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20. Describe five types of malpractices that the government protects consumers against by traders. (10mks)

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21. Outline five reasons why an ethical practice is necessary in product promotion. (10mks)

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22. Explain insurance policies that the owner of a supermarket may find useful for the business (10mks)

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END TERM 1-2020.

FORM 3

BUSINESS STUDIES PAPER 1

TIME: 2 HOURS.

NAME

ADM NUMBER

DATE.....

-Write your name, admission number and exam date on the spaces above.

-This paper consists of 6 printed pages. Check the question paper to ascertain that all the pages are printed as indicated and that no question is missing.

-Answer ALL the questions in the spaces provided.

-Number your work and make sure your work is neatly presented.

For examiner's use only

QUESTION	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
SCORE															

16	17	18	19	20	21	22	23	24	25

TOTAL SCORE

1. State four functions of management directed towards the attainment of objectives. (4mks)

2. Outline four reasons for collapse of small scale business units (4mks)

3. Highlight four disadvantages of newspapers as a means of advertising (4mks)

4. Give four benefits that Kenya would get if the government encourages the location of businesses in rural areas. (4mks)

5. State four factors that lead to business mergers. (4mks)

6. State four emerging issues in the banking industry. (4mks)

7. Highlight four reasons why a country would rely more on primary level of production. (4mks)

8. In the absence of a written partnership agreement, the provisions of the Partnership Act will apply. State **four** such provisions. (4 mks)

9. Describe **four** ways in which the government creates an enabling business environment. (4 mks)

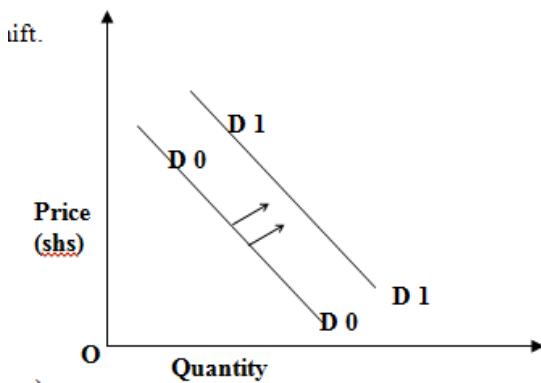
10. State **four** elements of communication process.

(4 mks)

11. Mention **four** circumstances under which an insurance company may re-insure.

(4mks)

12. The following diagram shows a shift in demand curve. Outline **four** reasons that can lead to this shift.(4mks)



13.State **four** characteristics of services.

(4mks)

14.Outline **four** disadvantages of division of labor.

(4mks)

15.Mention **four** advantages of using Automatic Vending Machines.

(4mks)

16. Give **four** benefits one gets from self-employment. (4mks)

17. Outline **four** factors to consider when intending to buy office furniture. (4mks)

18. Outline **four** principles of co-operatives. (4mks)

19. State **four** advantages of transporting oil by pipeline rather than by road. (4mks)

20. Outline **four** advantages of audio-visual communication. (4mks)

21. Highlight **four** ways in which the production activities of firms in your country affect the environment. (4mks)

22. Highlight **four** disadvantages of a private warehouse. (4mks)

23. Ragati and Nzonge operates a business with a stock valued at ksh.350,000 and have interests of $\frac{3}{5}$ and $\frac{2}{5}$ respectively. Ragati decided to insure $\frac{3}{4}$ of the stock against fire. Sometimes later fire occurred at their business

premises and destroyed stock worth ksh. 250,000. Calculate the amount of compensation Ragati would get from the insurance company. (4mks)

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24.Outline **four** sources of monopoly power. (4mks)

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25.Below are some factors that influence the operations of a business enterprise. Indicate with a (✓) tick in the appropriate column the business environment associated with each. (4mks)

Factors	Internal	External
Depreciation of Kenya shillings		
Drought		
Skilled managers		
Increase in raw materials		

NAME..... ADM NO..... CLASS.....

2020 FORM 3 TERM 1 OPENER EXAMS

CHEMISTRY.

FORM 3.

TIME: 2 HOURS.

INSTRUCTION TO CANDIDATES

Answer all the questions in the spaces provided.

FOR EXAMINERS USE ONLY.

QUESTIONS.	MAXIMUM SCORE.	CANDIDATE SCORE.
1-28.	100	

1. State two differences between luminous and non-luminous flames (2mks)

Non-luminous	Luminous

2. Name the best method that can be used to separate the following mixtures (3mks)
- (a) Sand and Water

(b) Ethanol and Water

(c) Sodium Chloride and Ammonium Chloride

3. Study the table below and use it to answer the questions that follow. (the letters do not represent the actual symbols of the elements.

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IIIC NUMBER					
NUMBER					

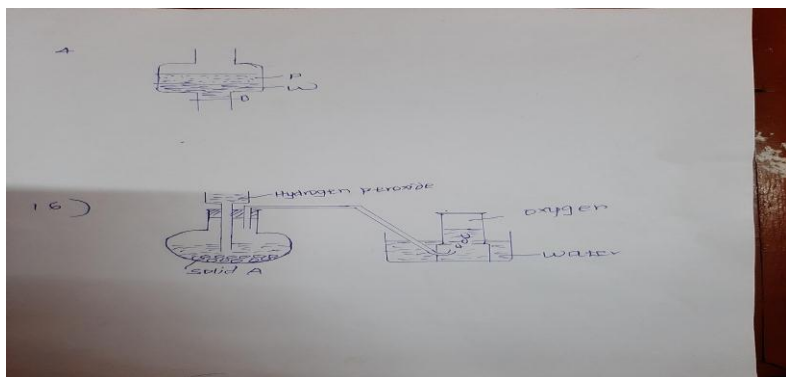
(a) Select two letters that represent the same element .Give a reason. (2mks)

(b) Give the number of neutrons in an atom of element D. (1mk)

(c) Identify an element that is unreactive. (1mk)

(d) Write the electron arrangement of B. (1mk)

4. A mixture of kerosene and water was shaken and left to separate as shown in the diagram below.



State the identity of; (2mks)

(i) P

(ii) W

5. (a) Complete and balance the following reaction. (1mk)



(b) State two uses of hydrogen gas. (2mks)

(c) Give a reason why nitric (V) acid is not used in the laboratory preparation of hydrogen gas.

(1mk)

6.(a) What are Isotopes?

(1mk)

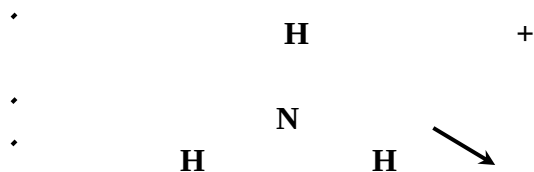
(b) The table below shows the relative atomic masses and percentage abundance of the isotopes M_1 and M_2 of element M.

ISOTOPES	RELATIVE ATOMIC MASS	% ABUNDANCE
M_1	60.57	59.71
M_2	62.83	40.29

Calculate the relative atomic mass of element M

(3mks)

7..The structure of an ammonium ion is shown above.



H

Name the type of bond represented in the diagram by:

(2mks)



8. When calcium granules are mixed with dilute sulphuric acid, a reaction is observed which soon stops. Explain giving the relevant equation. (2mks)

9. Below are pH values of some solutions.

TION				

a) Which solution is likely to be;

(i) Acidic rain

(1mk)

(ii) Sodium Chloride

(1mk)

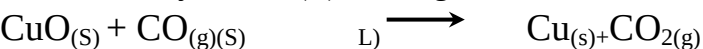
(iii) Dilute sulphuric VI acid

(1mk)

(iv) Sodium hydroxide acid

(1mk)

10. Dry carbon (II) Oxide gas reacts with heated copper (II) oxide as shown in the equation below.



(a) Name the process undergone by Copper (II) Oxide. Explain. (2mks)

(b) Name another gas that can be used to perform the same function as carbon (II) oxide gas in the above reaction. (1mk)

11. In the very cold countries, salts are sprinkled on the roads during winter. Explain why this is important.
(1mk)

12. Explain the meaning of each of the following terms;

(a) Efflorescence (1mk)

(b) Hygroscopy (1mk)

(c) Deliquescence (1mk)

13. Using dots (●) and crosses (X), show bonding in magnesium nitride (Mg=12, N=7).(2mks)

14. Below is a table of 1st ionization energies for elements A, B, C, D which are metals

ENTS				
ATION ENERGY (KJ/MOLE)				

Arrange the elements in order of increasing reactivity. (2mks)

15. Dilute sulphuric (VI) acid was added to a compound of magnesium P. The solid reacted with the acid to form a colourless solution Q and colourless gas R which formed a white precipitate when bubbled through lime water.

(a) Name;

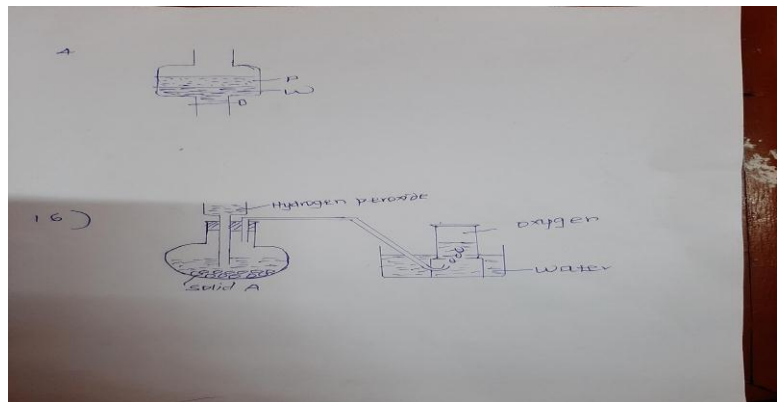
(i) Compound P (1mk)

(ii) Solution Q (1mk)

(iii) Colorless gas R (1mk)

(b) Write a chemical equation for the reaction that took place. (1mk)

16. The diagram below represents a set up that can be used to prepare and collect oxygen gas.



(a) Name solid A and state its role.

Solid A (1mk)

Role (1mk)

(b) What property of oxygen makes it possible for its collection as indicated by the diagram? (1mk)

(c) Explain why it is important not to collect any gas for the first few seconds of the experiment. (1mk)

(d) State two uses of oxygen gas. (2mks)

17. The chromatogram below was obtained from a contaminated food sample P. contaminants Q,R,S and T are suspected to be in P. Use it to answer the following questions.



(a) Identify the contaminants in mixture P. (1mk)

(b) Which is the least soluble contaminant in P? (1mk)

18. Two salts P and Q are weighed before and after heating. The results are given in the table below.

SALT	MASS IN GRAMS	
	BEFORE HEATING	AFTER HEATING
P		
Q		

(a) Which one is likely to be sodium carbonate? Explain. (2mks)

(b) If the other salt is zinc carbonate, state the observation made in the process of heating. (1mk)

19. A form two student was stung by a nettle plant and their chemistry teacher applied ammonia solution to the affected area of the skin and the student was relieved of pain. Explain. (2mks)

20. The table below gives some properties of substances I, II, III and IV. Study it and answer the questions that follow;

SUBSTANCE	PHYSICAL PROPERTIES	PHYSICAL CONDUCTIVITY	CHEMICAL PROPERTIES
I			
II			
III			
IV			

	D	EN		
	not conduct	cts		
	cts	cts		
	not conduct	not conduct		
	not conduct	not conduct		

(i) What type of bonding exists in substances

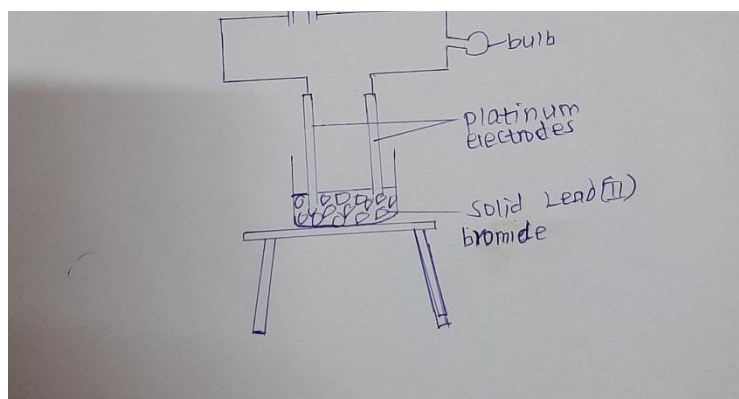
I (1mk)

II (1mk)

III (1mk)

(ii) Which substance is likely to be sulphur? Explain . (2mks)

21. In an experiment to investigate the conductivity of substances, a student used the setup shown below.



The student noted that the bulb did not light.

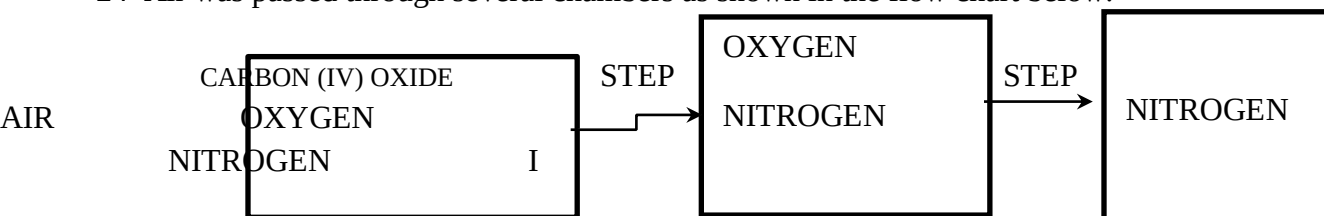
(a) What had been omitted in the set-up? (1mk)

(b) Explain why the bulb lights when the omission is corrected. (2mks)

22 In terms of structure and bonding, explain why graphite is used as a lubricant. (2mks)

23 State two uses of carbon (IV) oxide. (2mks)

24 Air was passed through several chambers as shown in the flow chart below.



(a) Name the gases removed in;

(i) Step I

(1mk)

(ii) Step II

(1mk)

b) State the reagent that can be used in :

(i) Step I

(1mk)

(ii) Step II

(1mk)

25 Name two conditions for rusting.

(1mk)

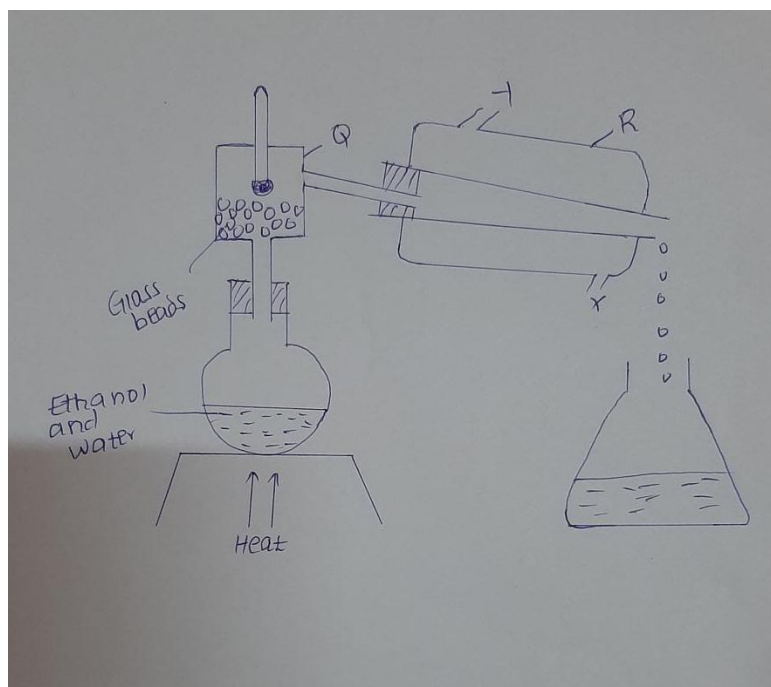
(b) State three methods that are used to prevent rusting.

(3mks)

26 State two uses of Sodium carbonate.

(2mks)

27 The diagram below shows a set-up of apparatus used to separate a mixture of ethanol (B.P = 78°C) and water (B.P = 100°C).



(a) Name the part labeled Q and state its function.

(2mks)

(b) Name the apparatus R.

(1mk)

(c) In the diagram, label water in and water out.

(2mks)

(d) Name the distillate that was collected first. Explain.

(2mks)

(e) Why is a round bottomed flask preferred for the experiment instead of the flat bottomed flask?
(1mk)

(f) What is the purpose of glass beads? (1)

28 The grid below is a section of the periodic table. The letters do not represent the actual symbols of the elements. Use it to answer the questions that follow.

U				S	
		V		W	X
Y					

(a) Select the elements which belong to the same;

(i) Group (1mk)

(ii) Period (1mk)

(b) What name is given to the family of elements to which X belongs? (1mk)

(c) Write the chemical formula of the carbonate of element Y. (1mk)

(d) Which letter represents the most reactive?

(i) Metal (1mk)

(ii) Non-metal (1mk)

(e) Name the bond formed when U and W react. Explain your answer. (2mks)

(f) Indicate on the grid the position of an element represented by letter R whose atomic number is 15.
(1mk)

(g) Identify one element which does not form an oxide. (1mk)

(h) Write three uses of alkali metals. (3mks)

Name..... Adm No.....

School.....Signature.....

Date.....

COMPUTER STUDIES

PAPER 1 (Theory) 451/1

TIME: 2HRS

END-TERM 1 FORM 3

Answer ALL the questions in the space provided (70 MARKS)

1. a) State two areas where computers are used. (2mks)

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.....

b) Give any two advantages of using computers in these areas (4mks)

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2. Give two advantages of a Digital Versatile Disk over a normal Compact disk. (4mks)

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.....

3. a) Name two types of buses found on the computer motherboard (2mks)

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.....
.....

b) State the purpose of each of the types of buses named in (a) above. (2mks)

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.....

4. Describe three mouse clicking techniques. (3mks)

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5. a) Explain the purpose of registers in a computer system (2mk)

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..... b) Name two multiprogramming operating systems (2mks)

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.....6. List two examples of;

i) Third generation languages (2mks)

ii) Object oriented languages (2mks)

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7. Give two types of storage devices used in a computer giving an example of each. (4mks)

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8. Give any three precautionary measures considered to ensure the safety of computers in a computer laboratory (3mks)

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9. As regards to communication within a computer network, what do you understand by the following terms: (4mks)

i) Point –to-point

ii) Broadcast

.....

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10. Give any two types of data entries that may be input into a worksheet cell (2mks)

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.....11. Explain the importance of using a Database management system for storage of files in an organization. (6mks)

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12. Distinguish between source program and object code in programming (2mks)

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.....13. a) State any **three** disadvantages of using computers (3mks)

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.....b) Give **three** examples of scanning devices (3mks)

c) Some computer system still use command line interfaces. Briefly explain four features of command line interface. (8mks)

14. Explain the following terms as used in Computer Studies

i) Firmware (2mks)

ii) Utilities (2mks)

..15. Briefly explain the functions of each of the following;

i) Ordinary Counter Register (OC) (2mks)

ii) Instruction Register (IR) (2mks)

iii) Accumulator Register (2mks)

2020 FORM 3 TERM 1 OPENER EXAMS

FORM 3 C.R.E.

TIME: 2 HOURS

NAME:.....ADM.NO:.....CLASS:.....

INSTRUCTIONS:

- 1. Answer all the questions in the spaces provided.
- 2. Answers must be written clearly.
- 3. Answer the questions in English language

1. a) State eight promises that God made to Abraham during his call. (8mks)

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b) Give the incidences when Abraham demonstrated his faith in God (7mks)

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c) State five modern forms of the covenant. (5mks)

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2. a) Outline 6 effects of idolatry in Israel during the time of Elijah (6mks)

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.....

b) Describe Mt. Carmel contest (1 kings 18:17-46) (8mks)

c) Identify six forms of idolatry that threatens the existence of Christianity today. (6mks)

3. a) In which ways was the birth of Jesus going to be extra-ordinary (6mks)

b) Describe the temptations of Jesus in the wilderness (8mks)

[illegible]

c) Identify the temptations that young people face in Kenya today. (6mks)

[illegible]

4. a) Relate the parable of the wicked tenants (Luke 20: 9-18) (8mks)

[illegible]

b) Name the methods used by Jesus to spread the good news (6mks)

c) State the reasons why people find it difficult to accept the gospel (6mks)

5. a) Name six specialists in traditional African communities. (6mks)

b) Give reasons why cleansing rituals were performed in Traditional African Society (7mks)

c) State the factors that undermine the roles of elders. (7mks)

Name..... Adm No.....

School.....Signature.....

Date.....

CHRISTIAN RELIGIOUS EDUCATION

PAPER 1 313/1

TIME: 2HRS

END-TERM 1 FORM 3

Answer FIVE questions in the answer booklet provided.

1. (a) Outline five books of Apocrypha (5mks)
(b) Identify seven responsibilities given to human beings by God in the Genesis stories of creation. (7mks)
(c) Explain the consequences of sin in relation to the Genesis stories of the fall of human beings. (8mks)
2. (a) Identify six roles of prophets in the Old Testament. (6 marks)
(b) Describe the teaching of prophet Amos on Israel's election. (6 marks)
(c) State eight factors that hinder Christians from practicing their faith in God. (8 marks)
3. a) State six ways in which the Israelites worshiped god in the wilderness during the exodus. (6mks)
b) State five reasons why Samuel was opposed to the idea of having a king in Israel. (5mks)
c) What factors have contributed to sexual immorality among the youth in Kenya today? (7mks)
4. a) Give reasons why the Northern Kingdom of Israel was destroyed before the Southern kingdom of Judah. (8mks)
b) State five practices of idolatry during the time of Prophet Elijah. (5mks)
c) Identify the effects of divorce on children in Kenya today. (7mks)
5. (a) Identify six occasions when sacrifices are made in Traditional African communities. (6mks)
(b) State seven roles of elders in traditional African communities. (7mks)
(c) Write down seven teachings about God from the traditional African myths of creation. (7mks)

NAME: _____ ADM NO: _____

DATE: _____ CLASS _____ SIGNATURE _____

2020 FORM 3 TERM 1 OPENER EXAMS

ENGLISH FORM 3

TIME 2 ½ HOURS

INSTRUCTIONS

1. Write your name and admission number on the spaces provided above
2. Answer all questions on the spaces provided
3. Confirm that the paper has all sections before beginning the exam

This paper consists of six sections as follows:

SECTION A	WRITING	20MKS
SECTION B	COMPREHENSION	20MKS
SECTION C	CLOZE TEST	10MKS
SECTION D	ORAL SKILLS	10MKS
SECTION E	LITERARY APPRECIATION	20MKS
SECTION F	GRAMMAR	20MKS
TOTAL		100MKS

SECTION A : WRITING (20MKS)

1. You are the chairperson of Amani Debate club in your school. Last Saturday, you had invited a neighbouring school for a debate in your school. The rest of the school also attended the debate; which ended at 6:00pm instead of 4:00pm as earlier agreed. This delayed other school programmes.
- a) Write an apology letter to the Principal of your school, citing reasons for the delay (14mks)

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the entire width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

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SECTION B: COMPREHENSION (20MKS)

Read the passage below and answer the questions that follow.

We are often advised to be confident: confidence comes when we feel passionately about what we perform, what we have achieved, and what we can offer.

Lack of confidence can hinder things, ranging from how you get ahead in your career to the number of skills you are able to acquire. It is a determining factor in the way you direct your career. Confidence is conveyed from the inner to the outer through the words you speak and the posture you assume. Building confidence is not a one day affair. Neither does it take other people to build your confidence.

I remember an interview I once conducted for secretaries, and the memory of one woman is intact. She stammered from the point she entered the office to the moment she left. Thinking that it was interview jitters, we called her again, based on her technical qualifications rather than her presentation. She presented the same problem. We settled on a different candidate because confidence was a key qualification for the job which involved dealing with clients and colleagues.

Not many entry-level and first-time job-seekers are coached or tutored on how to gain more self-confidence. Instead, we tell them to “attend a few events, meet more people and get the hang of it.” That is not the right approach. Often, by the time you “get the hang of it.” You will have probably ruined a few chances, stepped on a few toes, and generally spoilt a few opportunities. Confidence starts from within, which means working with yourself to find an effective method on your capabilities.

Having had quite some experience interviewing and being interviewed, I have developed three sure-fire ways that not only help me before an interview, but improve my general self-confidence. You must start with one as you progress and eventually build your own self confidence.

I have learnt that confidence begins with pushing yourself towards positive attitude. Before an interview, I tell myself that I am capable of great things, great performance and that interview is only a little chat I must have to get a job. Not only do I talk myself into entering the interview room, but I also talk myself into achieving success, whether or not I need the job about 90 percent of the time, I have been successful in interviews.

Doubting oneself only leads to criticizing oneself, which plants deep roots that can be almost impossible to weed out. My mentor always used to say, “You are what you think you are.”

Secondly, beginning with one small step rather than giant leaps will ensure that you tackle interviews and first jobs with a lot of self-confidence. There are two ways to go through this.

The first one has to do with something you that like doing, such as writing a song. However, this should not be just any song but a new one created out of your love for music. My personal pre-interview confidence boost is to cook something that I have never tried before. This boosts my self-confidence to new levels. After achieving that particular goal, you must pat yourself on the back.

The third step to building confidence involves holding onto that little surge of confidence. Once you have tackled what you have previously found to be difficult, then the next step would be to focus on building your self confidence a lot further.

Ensure that this becomes not only a monthly confidence boost, but a daily one because, as we are all programmed, we must build our self-confidence habitually to maintain it. Routines have been known to help anybody to acquire and maintain confidence.

From daily Nation 20th May 2013.

Questions

a) Define confidence according to the passage

(3mks)

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b) How is confidence conveyed?

(2mks)

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.....

c) What suggestions are given to first-time job seekers to help them in gaining more self – confidence?

(3mks)

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d) What does the writer identify as a major hindrance to confidence?

(1mk)

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e) According to the writer, what two key qualifications were they looking for in interview for secretaries?

(2mks)

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f) Identify the **three** ways the writer mentions that help before an interview and improve in self confidence. (3mks)

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g) In **note** form, write what the writer has been doing to achieve success in interview (4mks)

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h) Explain the meaning of the following words as used in the passage (2mks)

- i. Interview jitter.....
-
- ii. Habitually
-

SECTION C : CLOZE TEST (10MKS)

Fill in the blank spaces with the most appropriate words.

Some parents prefer day schools (1) _____ boarding schools for their children. They say that boarding schools do not give them a (2) _____ to know the teachers

of their children (3)_____. These parents also feel that boarding schools lock them (4)_____ of their children’s learning. This is (5)_____some boarding schools (6) _____ parents to visit their children (7) _____ once a term. They say (8) _____ a day school provides (9)_____ time for parents to monitor (10)_____ children’s learning.

SECTION D: ORAL SKILLS (10MKS)

a) The following words have either /s/ or /z/ sound. Group them correctly (5mks)

- Rise

Presiden

t

lose

advise

preceden

t
- Loose

advice

clause
- rice

lice

/s/	/z/

b) For each of the following words, underline the stressed syllable

(5mks)

- i. Mis.use (noun)
- ii. re.gis.ter (verb)
- iii. a.larm (noun)
- iv. a.gree.ment (noun)
- v. ap.proach (noun)

SECTION E: LITERARY APPRECIATION

(20MKS)

Read the narrative below and answer the questions that follow

Once upon a time there lived a very beautiful girl known as Karia in a distant village. One day, Karia and her mother went to dig in their garden by the edge of the forest that was home to ogres that could talk and even sing. When it was time to leave, the mother called out to her daughter who was bathing in the stream near the garden, ‘Karia, we had better rush home; darkness will soon descend upon us and there are many ogres along the way’.

Karia hurriedly picked up the luggage her mother had packed and they went home. On reaching home, Karia realized that she had left her beautiful necklace by the stream where she had taken a bath. Her grandmother had given it to her saying, ‘Take good care of this necklace and pass it on to your eldest daughter.’

Karia was very depressed at the fact that she had left it beside the stream.

‘Mother, I have to go back for the necklace. I forgot it by the stream,’ Karia said. ‘No my dear daughter, it is already dark outside and besides the ogres are roaming the forest.’ She could not bear the thought of her only daughter setting out in the pitch of darkness.

The girl started crying, saying that if she waited for daylight, she would find the necklace missing. Her mother withdrew to the kitchen to check whether the food was ready, she came out only to find Karia missing.

“Karia! Karia! Karia! Where are you? Come!” She called out. It dawned on her that Karia must have gone back to the stream for the necklace. She immediately ran to her brother-in-law’s home in search of her husband panting; ‘Father of Karia, gather other men, Karia has gone back to the stream for her necklace! Oh, my only daughter...those ogres...what will I do? ‘What is wrong with my daughter? I could have bought her another necklace,’ said Karia’s father.

It took about an hour for the men to prepare themselves and set out for the search. Their hearts beat with apprehension as they pressed through the dark.

Karia had been running swiftly and by this time she was by the edge of the forest. She slowed down for the remaining part of the journey in order to catch her breath. The early morning wind chilled her skin. The thought, ‘If I failed to get back the necklace, what will I tell my granny?’ quickened her pace to where she had undressed and found that her necklace was intact. Joy flooded her heart. She quickly picked up the necklace and started on her way home with a sprinty leap, laughing at her mother’s fear of ogres.

She had hardly taken a few steps, when she met an ogre squarely planted at the middle of the path. In order to be allowed to pass she sang:

You ogre
Do not bother me
I am Njanwa’s daughter
Take this bead
And let me pass

The ogre accepted the bead and let her pass only to run through the bush past her and stop in the middle of the road. She reached the point where it stood blocking the way and then she repeated her song. By the time she was half way back home, she had run out of beads.

Having no alternative, she tearfully said,

‘Take this arm and let me pass.’

Through her sorrow and pain, she thought of the necklace and the ogre that still demanded something in exchange for passage.

After a while, she had only one leg and so the ogre decided to eat her up. Her father and the village elders came upon the satisfied ogre. They speared it and slit it open its belly. To their delight, Karia emerged, holding the beautiful necklace. There ends the story.

(Adapted from Narratives from Africa and India for secondary schools and colleges)

Questions

a) Classify the narrative above (2mks)

b) State any two functions of each of the formulae.

i. “Once upon a time...” (2mks)

ii. “There ends my story.” (2mks)

c) Apart from the formulae “Once upon a time...” and “there ends my story”, identify and illustrate three features that indicate that the story is an oral narrative. (6mks)

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d) Identify and illustrate two character traits of Karia. (4mks)

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e) Identify and illustrate one theme evident in the narrative above (2mks)

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.....

f) With illustration, identify one economic activity that is practiced by the community in the narrative

(2mks)

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.....

SECTION F: GRAMMAR (20MKS)

a) Fill in the blank spaces using the most appropriate form of the word in the brackets (4mks)

- i. It was never his _____ to start a fight. (intend)
- ii. We all got angry because of the speaker's _____ remarks. (offend)
- iii. He is an _____ expert. (electron)
- iv. He _____ believes in hard work. (true)

b) Join the pairs of sentences into one sentence using the conjunction provided (3mks)

i. -He has never done any fighting.

-He is a soldier.(although)

.....

.....

ii. -Everybody else in that country was blind.

-The one-eyed man became a king. (because)

.....

.....

iii. -I met the man.

-The man's wife bought a car. (whose)

.....

.....

c) Rewrite the sentences as instructed. Do not change the meaning.

(3mks)

i. I will only go if he asks me to. (Begin ; unless)

.....

.....

ii. I think we are being followed by somebody. (change into active voice)

.....

.....

iii. "Are you writing the proposal?" asked the supervisor. (Change into reported speech)

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d) Fill in the blank spaces using the correct preposition

(4mks)

i. The visitors arrived_____ Christmas day

ii. The explanation you want is_____ the top of page 20.

iii. The student insisted _____ doing the test at night.

iv. There is no possibility _____finishing this work today.

e) Use the simple present tense of the verb in brackets to fill in the blank spaces (3mks)

- i. Each of the books_____ (cost) over three hundred shillings.
- ii. Neither of the parcels_____(have) what I was expecting.
- iii. Few people_____(believe) in superstitions.

g) State the type of phrase underlined. (3mks)

e.g The girl from our village won the award- noun phrase

- i. The dog might eat the cake

.....

- ii. We are expecting him to come next year.

.....

- iii. Swimming costumes are very expensive

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ENGLISH 101/2**END TERM 1****FORM THREE****TERM ONE 2020**

Name.....Adm No.....Class.....

Instructions to candidates

- Answer ALL the questions

For Examiners Use Only

Question	Maximum score	Candidate's score
1.Comprehension	20	
2.Literary Appreciation	25	
3.Poetry	20	
4.Grammar	15	
Total score	80	

1. Read the passage below and answer the questions that follow (20mks)

WORRY has completely defeated me. My mind was so confused and troubled that I could see no joy in living. My nerves were so strained that I could neither sleep at night nor relax by day. My three young children were widely separated, living with relatives. My husband was in another city trying to establish a law practice. I felt all the insecurities and uncertainties of the post –war readjustment period.

I was threatening my husband's career, my children natural endowment of a happy, normal home life, and I was also threatening my own life. My husband could find no housing, and the only solution was to build. Everything depended on my getting well. The more I realized this and the harder I would try; the greater would my fear of failure. Then I developed a fear of planning for any responsibility. I felt that I could no longer trust myself. I felt I was a complete failure.

When all was darkest and there seemed to be no help, my mother did something for me that I will never forget or ceases being grateful for. She shocked me into fighting back.

She upbraided me for giving in and for losing control of my nerves and my mind, she challenged me to get up out of bed and fight for all I had. She said I was giving in to the situation, fearing it instead of facing it, running from life instead of living it.

So I did start fighting from that day on. That very weekend I told my parents they could go home, because I was going to take over; and I did what seemed impossible at the time. I was left alone to care for my two younger children. I slept well. I began to eat better, and my spirits began to improve.

A week later when they returned to visit me again, they found me singing at my ironing. I had a sense of well-being because I had begun to fight a battle and I was winning. I shall never forget this lesson..... if a situation seems insurmountable, face it! Start fighting! Don't give in.

From that time on I forced myself to work, and lost myself to work. Finally I gathered my children together and joined my husband in our new home. I resolved that I would become well enough to give my family a strong, happy mother. I became engrossed with plans for my children, plans for my husband, and plans for everything except for me. I became too busy to think of myself. And it was that the real miracle happened.

I grew stronger and stronger and could wake up with the joy of well - being, the joy of planning for a new day ahead, the joy of living. And although day of depression did creep in occasionally after that, especially when I was tired, I would tell myself not to think or try to reason with myself on those days and gradually they became fewer and fewer and finally disappeared.

Now, a year later, I have a very happy, successful husband, a beautiful home that I can work in sixteen hours a day, and three healthy, happy children and for myself , peace of mind.

(Adapted from "How to stop worrying and start living")

Questions

(a) Give **five** effects of worry on the writer.

(5mks)

(b) What do you think was the cause of the writer's worry? (2mks)

(c) Describe the element of **irony** in the writer's life. (3mks)

(d) How does the writer get out of this terrible situation? (2mks)

(e) What **steps** does the writer undertake to overcome the problem? (4mks)

(f) In a paragraph of about **60 words** explain the benefits of the writer's efforts to overcome worry. (4mks)

2. Read the following excerpt and answer the questions that follow

However, the notion that he was about to hand over his own daughter to a gangster continued to **gnaw** at the conscience of Ole Kaelo relentlessly. He felt guilty, especially when he recalled the atrocities that were known to have been committed by Oloisudori over the years. But another voice told him quietly that he was being foolish and

unreasonable to question his own conscience over the matter of Oloisudori, for he was just one among many who were enjoying the fruits of their labour. And it was hardly anybody's business to know how honest that labour was. After all, the small voice reassured him tauntingly, those who committed bigger crimes such as Goldenberg and Anglo-leasing, were still enjoying the 'fruits of their labour.' Had they not invested the yields of their ill-gotten money in housing estates, in shares, in import and exports in tourism, in transport and in other trades, just as Oloisudori had done?

When he went to bed later that evening, he remained awake for many hours **pondering** over those disturbing thoughts that went through his mind fleetingly, like water that churned violently in a turbulent sea. He thought of Oloisudori's **impending visit** and his intended marriage to Resian. He knew the success or failure of the event would determine the fate of his business. Even his continued ownership of that house where he and his family lived, depended on the outcomes of that event. Should Oloisudori fail to get Resian and recall the loan he had extended to him to buy that house, **he was done**. And knowing Oloisudori, he could very easily draw the rug from beneath his feet, leaving him vulnerable to all kinds of vagaries. And the thoughts gave him anxious moments.

At dawn when sleep overtook him, Ole Kaelo had a pleasant dream. Resian had consented to Oloisudori's proposal. After Oloisudori reported that to him, he was greatly pleased and relieved. His wife was rapturous. Although they were astonished at the turn of events, they were relieved to know that they would not have to live with the guilt of having forced their daughter to get married. What a wise child his once hardheaded daughter had turned to be after all! And how devious! After all those years of sullenness, awkwardness and tactlessness, she had finally brought relief to their life and ushered in a period of peace and tranquility. But then, it was just that. A dream!

Questions

a).Place the excerpt in its **immediate** context. (4mks)

b).Discuss three **major issues** in this excerpt. (6mks)

c).Discuss two character **traits** of ole kaelo in this excerpt. (4mks)

d). “and the thoughts gave him anxious moments” (**add a question tag.**) (1mk)

e).Discuss any **three aspects** of **style** in this excerpt. (6mks)

f).Explain the **meaning** of the following expressions from the excerpt. (4mks)

- (i). Gnaw.....
- (ii). Pondering.....
- (iii). Impending visit
- (iv). He was done.....

3. Read the following *oral poem* and then answer the questions which follow

O elephant possessor of a saving- basket full of money

O elephant, huge as a hill, even in a crouching posture,

O elephant, enfolded by honour: demon, flapping fans of war.

Demon who snaps tree branches into many pieces and moves on the forest farm,

O elephant, whom ignores ‘ I have fled to my father for refuge’.

Let alone ‘to my mother’ Mountains Animal, Huge

Best who tears a man like a garment and hangs him up on a tree,

The sight of whom causes people to stampede towards a hill of safety,

My chant is a salute to the elephant, Ajanaku who walks with a heavy tread,

Demon who swallows palm-fruit bunches, whole, even with the spiky pistil –cells,

O elephant, praise named, Laaye, massive animal blackish –grey in complexion,

O elephant, who single-handedly cause a tremor in a dense tropical forest,

O elephant, who stands sturdy and alert, who walks slowly as if reluctantly,

O elephant, whom one sees and points towards with all one's fingers.

The hunters boast at home is not repeated when he really meets the elephant.

The hunters boast at home is not repeated before the elephant,

Ajanaku looks back with difficulty like a person suffering from a sprained neck.

The elephant has a porter's – knot without having any load on his head.

The elephant's head is his burden which he balances,

O elephant, praise named, Laaye, O death, please stop following me.

This part and parcel of the elephant's appellation.

If you wish to know the elephant, the elephant who is a veritable ferry –man.

The elephant whom honuor matches, the elephant who continually swings his trunk.

His upper fly-switch, It's the elephant whose eyes are veritable water –jars,

O elephant, the vagrant par excellence.

Whose molar teeth are as wide as palm- oil pits in Ijesaland,

O elephant, lord of the forest, respectfully called Oriiribobo

O elephant whose teeth are like shafts, One tooth of his a porter's load,

O elephant fondly called Otiko, Who has a beast- of burden's proper nock,

O elephant, whom the hunter at other times aces face to face.

O elephant, whom the hunters at other times seas from the rear.

Beast who carries mortars and yet walks with a swaggering gaint.

Primeval leper, animal treading ponderously.

Questions

(a) What **type of** oral poem is this? (2mks)

(b) Identify, illustrate and give the effect of any **three** features of style used in the poem.(6mks)

(c) How is the elephant **portrayed** in the poem? (2mks)

(d) What is the **persona's attitude** towards the elephant? (2mks)

(e) Identify any **two** economic activities of the community from which this poem is derived. (4mks)

(f) Identify the main **theme** of this poem. (2mks)

(g) Explain any **two** problems you may encounter when collecting material for this genre. (2mks)

4. **GRAMMAR**

a). **Rewrite** the following sentences as instructed (2mks)

1. My examination results were released only after I had cleared the fee balance. **(Begin: Not until.....)**

2. If I were the Minister for National Security, I would ensure tighter security checks at the borders.

(Begin: Were I)

3. They had handled the case so carelessly that we were not satisfied. **(Begin : So ...)**

4. "I shall expect to see you next Wednesday," the teacher told the student. **(Write in reported speech.)**

b). Use the **correct form** of the word in brackets in the sentences that follow. (3mks)

i. The modern world today has manyforms of communication (**sophistication**)

ii The gas was stored in acontainer (**pressure**)

iii. Do not wait for the bus, it comes to this town very (regular).

c). Give the **meaning** of the underlined **idiomatic expression**. (2mks)

(i) The teacher asked him to stop beating about the bush.

(ii) The politicians asked them to cross the bridge when they reach it.

d). Choose the correct **pronoun** in the following sentences. (2mks)

- (i) She knew all about my friend and _____(I/me)
(ii) What would you do if you were _____? (she/her)

e).Complete each blank space in the following sentences with **appropriate choice** from:
(*few, a few, little ,a little, some*)

(2mks)

- (i) If there is tea left in that flask, I would like to have _____
(ii) Since animal proteins are very expensive _____people eat enough

f) .Replace the underlined word with the correct **phrasal verb**. (2marks)

1. The class teacher promised to solve the problem.(sort)
2. He tried all his dirty tricks on us but he did not succeed .(come)
3. The nurse couldn't tolerate the noise(put)
4. The mayor assumed that the problem had been got rid of.(iron)

g).Rewrite the sentences below as to remove **gender bias**. (2mks)

1. The headmistress advised her girls to keep off drugs.
2. The father left his sons and daughters a big estate.

NAME:.....ADM NO.....

GEOGRAPHY FORM 3

2020 FORM 3 TERM 1 OPENER EXAMS

INSTRUCTIONS TO STUDENTS:

1. Write your name, admission number and class in the spaces provided.
2. Answer all the questions in this paper.
3. All answer must be written in the spaces provided.
4. Candidates should check the question paper to ascertain that all the pages are printed as indicated and that no questions are missing.

SECTION A(25MKS)- ANSWER ALL THE QUESTIONS IN THIS SECTION (25MKS)

1. (a) State three reasons why the interior of the earth has high temperatures. 3mks

- (b) Give two differences between Sial and Sima. 2mks

2. (a) Give two ways in which heat is transferred to the atmosphere. 2mks

- (b) Describe how humidity is measured in a weather station. 4mks

3. (a) State two conditions that are necessary for the formation of petroleum. 2mks

(b) The table below shows some examples of rocks. Complete it.

3mks

<u>Original rock</u>	<u>metamorphic rock</u>
Granite	
Sand	
Clay	

4. (a) Name two effects of horizontal earth movement .

2mks

(b) State two weaknesses of continental drift theory.

2mks

5. (a) Name two modern methods used in weather forecasting .

2mks

(b) State three ways in which weather forecasting is important to man.

3mks

SECTION B(ANSWER ALL THE QUESTIONS (75MKS

6. (a) (i) what is folding.

2mks

(ii) State three factors which influence folding. 3mks

(iii) Differentiate between symmetrical and asymmetrical fold. 2mks

(b)(i) Name three fold mountains found outside Africa. 3mks

(ii) Apart from fold mountains, name two features produced by folding. 2mks

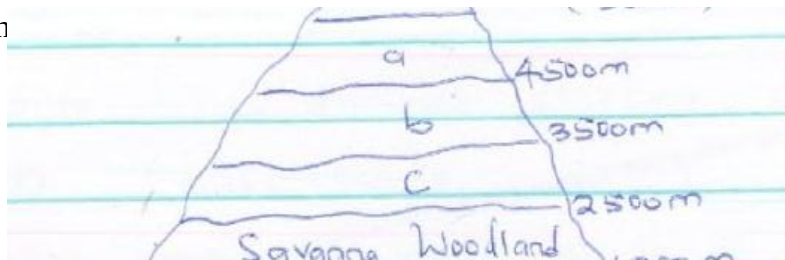
(iii) Describe how fold mountains are formed. 5mks

(c) Explain four ways in which fold mountains influence human activities. 8mks

7. (a)(i) Name two types of vegetation. 2mks

(ii) Give two Pssyigraphic factors influencing vegetation distribution . 2mks

(b) The diagram below sh answer the questions that



follow

(i) Name the zones marked a,b,c and d . 4mks

(ii) Give one reason why there are no vegetation cover on top of the mountain. 1mk

(c)(i) State the name given to temperate grassland vegetation in :2mks

-North America-

-Argentina-

(ii) Explain three uses of Savanna vegetation. 6mks

d) Explain four characteristics of tropical desert vegetation. 8mks

8. (a) The table below shows the production of various horticultural crops in Kenya between 2008 and 2010 in “000” tonnes.

Year			
2008			
2009			
2010			
Crop			
Tomatoes			

(i) *Using a scale of 1cm represent 50,000 tonnes, draw a comparative bar graph to present the data above.* 8mks

(ii) Name the crop that has the highest production within the period . 1mk

(iii) Give two advantages of using comparative bar graphs. 4mks

b) identify the two types of photographs. 2mks

(c) Give two uses of photographs. 4mks

(d) Give two differences between aerial photographs and ground photographs. 4mks

(e) Define the following term used in photographs:

Dead ground

2mks

GEOGRAPHY 312/2

END TERM

FORM THREE

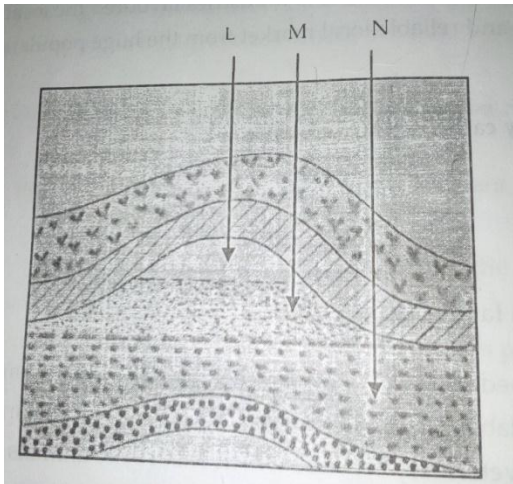
TERM ONE 2020

NAME: ADM NO: CLASS:

SECTION A

Answer ALL the questions in this section.

1. (a) Distinguish between forest and forestry. (2mks)
(b) State three problems facing forestry in Kenya. (3mks)
2. (a) Define economic Geography. (2mks)
(b) Give three natural economic factors of production exploited by man. (3mks)
3. (a) Give two hardwood species in Kenya (2mks)
(b) Give three characteristics of softwood in Canada. (3mks)
4. The diagram below shows the occurrence of petroleum in the earth's crust. Use it to answer questions (a) and (b)



- (a) Name the substances in the area labeled L, M and N. (3 mks)
- (b) Give two by-products obtained from the refining of crude oil (2 mks)

5. (a) Name two areas in South Africa where diamonds are mined. (2mks)
 (b) Identify three methods used in diamond mining in South Africa. (3 mks)

SECTION B

Answer questions six and any other two.

6. The table below gives the figures for the 1999 population census per district for central province of Kenya.

District	Population
Nyandarua	479902
Nyeri	661156
Kirinyaga	457105
Muranga	348304
Kiambu	744010
Thika	645713
Maragua	387969

- (a)(i) Beside bar graphs, name three other methods that can be used to represent the data above. (3mks)
- (ii) Draw a simple bar graph to represent the data in the table above. Use a scale of 1cm to represent 100,000 people. (6mks)
- (iii) State four advantages of using bar graphs to represent statistical of three densely populated districts. (4mks)
- (b) Calculate the total population of three densely populated districts. (2mks)
- (c) Explain three types of ground photograph (6mks)
- (d) State four advantages of a comparative line graph. (4mks)
7. (a) (i) Distinguish between indigenous and exotic forests. (2mks)
- (ii) Show how natural forest differ from planted forests (8mks)
- (b) Explain three factors that influence the distribution of softwood forests in Kenya (6mks)
- (c) (i) State three measures that are being taken in Kenya to conserve forests. (3mks)
- (ii) Explain three factors favouring the exploitation of softwood in Canada. (6mks)

8. (a) Explain five factors to consider before undertaking mining. (10mks)
- (b) Give three problems facing soda ash mining in Magadi (6mks)
- (c) (i) State three methods of placer/ alluvial mining. (3mks)
- (ii) Explain three negative effects of open cast mining. (6mks)
9. (a) Explain three reasons why only a small part of Kenya is forested (6mks)
- (b) Explain five ways in which diamond contributes to the economy of South Africa. (10mks)
- (c) (i) Name three leading minerals which are exploited in Kenya. (3mks)
- (d) Form four students conducted a field, study in mining in their local area.
- (i) State three methods of data presentation they were likely to use. (3mks)
- (ii) State three importance of work schedule during their fieldwork. (3mks)
10. (a) Differentiate between human geography and economic geography. (2mks)
- (b) Name at least FIVE human activities in Kenya. (5mks)
- (c) Your Geography class organized a field study on wildlife at Lake Nakuru National Park.
- (i) Apart from preparing a route map, identify other ways in which the class prepared for the study. (5mks)
- (ii) Why was it necessary to prepare a route map?(4mks)
- (i) Identify the activities the class was involved in during the study.(6mks)
- (ii) Give THREE methods the class used to record data.(3mks)

2020 FORM 3 TERM 1 OPENER EXAMS

FORM II HISTORY

TIME 2HRS

NAME: ADM NO: CLASS:

SECTION A (25MKS)

ANSWER ALL THE QUESTIONS IN THIS SECTION.

1. Name two sources of history of the Kenyan communities during the pre-colonial period. (2 Marks)

2. Give two factors which led to the migration and settlement of the Iteso into Kenya by 1800 A.D. (2 Marks)

3. Identify two aspects of the culture of the early man that had their origin in the late Stone Age. (2 Marks)

4. Identify two factors which favoured the development of crop growing in ancient Egypt. (2 Marks)

5. In what two ways did the Kenyan communities interacted during the pre-colonial period. (2 Marks)

6. What was the base of the political organization among the Cushites during the pre-colonial period. (1 mark)

7. Identify the two economic activities which the Maasai acquired as a result of interacting with the Agikuyu during their settlement in Kenya. (2 Marks)

8. What religious functions did the Orkoiyot of the Nandi perform during the pre-colonial period.? (2 Marks)

9. Name two Bantu communities in Kenya whose ancestors settled in the Mt., elgon area before migrating to their present homeland. (2 Marks)
10. State any two Arab families that ruled the coastal town. (2 Marks)
11. Distinguish between barter and currency trade. (2 Marks)
12. State two roles played by the Tuaregs in the Trans-Saharan Trade. . (2 Marks)
13. State two ways in which direct democracy is exercised in Kenya. . (2 Marks)

SECTION B

(45 MARKS)

ANSWER ANY THREE QUESTIONS FROM THIS SECTION.

14. (a) State any three theories about the origin of human beings. . (3 Marks)

(b) Discuss the benefits of the discovery of fire by early man. (12 Marks)

15. (a) State reasons for the migrations of Eastern Bantu speakers into Kenya by 1900A.D. (5 Marks)

b) Discuss the results of the migration and settlement of the Eastern Bantu into Kenya by 1900 A.D.
(10 marks)

16. (a) Why were the Portuguese able to control the Kenyan Coast between 1500 and 1700 A.D. (3 Marks)

(b) Describe the results of the Portuguese rule along the Kenyan Coast. (12 Marks)

17. (a) What were the economic activities of the people of the Asante Empire? (3 Marks)

(b) Describe the political organization of the Asante Empire during the 10th Century (12 Marks)

SECTION C

ANSWER ALL THE QUESTIONS IN THIS SECTION.

18. (a) State five values of good citizenship. . (5 Marks)

(b) Explain five factors that promote national unity in Kenya. (10 Marks)

19. b) Mention three stages of the constitution making process in Kenya. (3 Marks)

(a) Discuss six features of the independence constitution of Kenya? (12Marks)

Name..... Adm No.....
School.....Signature.....
Date.....

HISTORY AND GOVERNMENT

PAPER 1 311/1

TIME: 2HRS

END-TERM 1 FORM 3

SECTION A (20 marks)

Answer ALL the questions in this section in the answer booklet provided.

1. Identify two sources of Kenyan history (2mks)
2. Name one community in Kenya that belongs to the River Lake Nilotes (1mk)
3. State two economic activities of the Abagusii in Kenya during the pre- colonial period (2mks)
4. Give two reasons which led to the decline of Gedi during the 15th century (2mks)
5. Name the type of constitution used in Kenya (1mk)
6. Identify two peaceful ways of resolving conflicts (2mks)
7. Apart from the Nandi, name two other communities that resisted the establishment of colonial rule in Kenya. (2mks)
8. State the main duty of the Governor during the British colonial rule in Kenya (1mk)
9. State one reason why the colonial government established Local Native Councils in Kenya in 1924. (2mks)
10. Identify two features of African farming in Kenya during the colonial period (2mks)
11. Name one Ex- officio member of parliament in Kenya? (1mk)
12. Identify two negative consequences of urbanization in Kenya during the colonial period. (2 marks)

SECTION B (60 MARKS)

Answer ALL questions from this section in the answer booklet provided

- 13 (a) Why did the highlands Nilotes migrate from their original homeland during the pre-colonial period? (5mks)
- (b) Explain five results of the migration and settlement of the highland Nilotes in Kenya (10mks)
14. (a) What were the activities of the imperial British East Africa company (IBEA Co.) between 1888 and 1895? (5mks)
- (b) Why did imperial British East Africa Company (IBEA co.) rule come to an end in 1895? (10mks)
15. a) What economic reasons encouraged the British to colonize Kenya during 19th Century. (3mks)
- b) Explain six ways through which the colonial government promoted settler farming.. (12mks)
16. (a) State three ways through which communities in Kenya interacted during the pre- colonial period. (3mks)
- (b) Describe the social organization of the Agikuyu during the pre- colonial period (12mks)

JINANAMBARIDARASA.....

2020 FORM 3 TERM 1 OPENER EXAMS

KIDATO CHA TATU

MUDA: SAA 2 ½

MAAGIZO

- (a) Andika jina lako na nambari yako ya mtihani katika nafasi ulizoachiwa hapo juu.
- (b) Jibu maswali yote.
- (c) Majibu yote yaandikwe katika nafasi ulizoachiwa katika kijitabu hiki cha maswali.
- (d) Majibu yote lazima yaandikwe kwa lugha ya Kiswahili.
- (e) Watahiniwa lazima wahakikishe kwamba kurasa zote za karatasi hii zimepigwa chapa sawasawa na kuwa maswali yote yamo.

Kwa matumizi ya mtahini pekee.

Swali	Upeo	Alama
A	20	
B	15	
C	40	
D	10	
E	15	

[illegible]

kubishiana chemchemi za maji na mali asli. Asilimia kubwa ya mito imekauka, chemchemi za maji zinapotea na vinamisi vya milima ya taifa havipo tena kutokana na uharibifu wa misitu na utumiaji mbaya wa malisasili. Ndiyo maana hutajasita kukariri kuwa mabadiliko ya hali ya anga ni mtikisiko wa kitaifa kwani ni hali ya hatari na kuwa kila mtu ana jukumu kujitahidi kiasi cha uwezo wake kuchangia uhifadhi wa mazingira .Mvua ikinyesha tupande miti, tuhifadhi chemchemi za maji na sehemu za unyevunyevu tulinde mito na maziwa yetu;tulime huku tukihifadhi udongo wetu,tufuge ,si kwa haja ya kuwa na mifugo wengi bali kwa kukidhi mahitaji na pia kulinda mazingira;tujitayarishe kuhifadhi maji ya mvua kwa ajili ya matumizi ya nyumbani na kilimo tuzidishe juhudi za kupanda mimea mbalimbali ya chakula huku pia tukifufua ile ya kienyeji.

Hatakatika za kupambana kuhifadhi msitu wa Mau,ambao ni chanzo kikubwa zaidi cha maji na mito nchini na hata Afrika Mashariki zinajulikana .Kutokana na vitendo vya ulafi wa viongozi wetu msitu huu umevamiwa na kuharibiwa na hata unatishia maisha ya mito, maziwa, mbuga za kuhifadhi wanyama pori na mamilioni ya watu wanaotegemea msitu huu kwa njia moja au nyingine.Tunatambua kwamba juhudi nyingi zinafanywa na serikali kutokana na shinikizo za wa wakereketwa na wadau wa mazingira .Lakini lazima serikali itumie uwezo wake wote kulinda misitu ya Mau,Mlima Kenya,mlima Elgon na misitu mingine kwa jumla.Tusipuuze misitu na mazingira kwa sababu ya siasa duni.

Serikali haifai kuachiwajukumu la kulinda misitu pekee bali wananchi pia wanafaa wajitume kwani mazingira yanahusu uhai wa kila mmoja wetu.Kuntu umoja ni nguvu utengano ni udhaifu.Ndiyo maana tunachukua fursa hii kuwapongeza wakazi wa Malava katika kauti ya Kakamega ambao hivi majuzi walijitokeza mzomzo kupinga uharibifu wa msitu huo na kuungana pamoja kupanda takriban miche hamsini elfu.watu waharibifu wasiruhusiwe kuangamiza misitu , mazingira, mandhari na masiha kule Malava.

Serikali pia inapaswa kutenga hela kulinda misitu inayofunika milima.Milima ya taifa ni mandhari yanayopendeza nchini na lazima kila juhudi za kulinda uhai katika milima hii zifanye .Ni jukumu la maafisa wa ardhi ,wa mazingira na wa misitu kutoa ushauri kwa serikali kwamba kuna hatari ya misitu kumalizwa na watu wabinafsi.Sheria ihimizwe na wananchi wahamasishwe huhusu umuhimu wa kulinda na kuimarisha mazingira.

Maswali

- a. Toa kichwa mwafaka kwa taarifa hii. Al 1

- b. Mabadiliko ya hali ya anga husababishwa na nini? Al 2

- c. Toa athari tatu tatu zinazosababishwa na kubadilika kwa hali ya anga. Al 3

- d. Kwa nini mwandishi ansema hatasita kukariri kuhusu hali ya hatari? Al 3

e. Ni sababu gani zilichangia kuharibiwa kwa msitu wa Mau? Al 1

f. Eleza sababu za mwandishi kusema umoja ni nguvu utengano ni udhaifu. Al 1

g. Nini kiini cha kuzuia uharibifu wa msitu wa Malava? Al 2

h. Eleza maana ya vifungu vifuatayo kama vilivyotumiwa katika taarifa hii. Al 2

i. Vizazi vya kesho

ii. Mtikisiko wa taifa

D. MATUMIZI YA LUGHA

1. Andika neno lenye sauti zifuatazo . al 2

Kimadende, irabu ya nyuma juu, kitambaza, irabu ya kati chini.

2. (a) Mofimu ni nini? Al 2

(b) Ainisha mofimu katika neno lifuatalo. aingiapo. Al 2

3. Andika neno lenye muundo ufuatao.

KKIKKKI AL 2

4. Tumia kirejeshi cha mwisho kuandika sentensi ifuatyo.
Watoto wanaochezea pale ni wangu upya. Al 2

5. Tunga sentensi mbili kutofautisha kitenzi chuma. Al 2

6. Maneno yafuatayo yamo katika ngeli zipi (al 2)

i. Karatasi

ii. Nyangumi

7. Onyesha aina za maneno yaliyopigiwa mstari

i. Mwalimu msichana alienda Mombasa. Al 2

ii. Alichezewa na watoto wabaya sana.

8. Kirai ni nini? Al 2

9. Geuza sentensi ifutayo iwe katika hali ya ukubwa wingi. Al 2
Mtoto alimkamata mbuzi mwingine .

10. Andika kinyume cha: al 2
Jana walianika nguo za mvulana huyo.

11. Unda nomino kutokana na maneno yafuatayo. Al 2
i. Mwalimu

ii. Mwandishi

12. Yakinisha al 2
Wasiposoma kwa bidii hawatapita mtihani huo.

13. Pambanua sentensi ifuatayo kwa kielezo cha mstari. Al 2
Huyo ni mtoto mzuri sana.

14. “Tutaanza kulima kesho,” ,mama alimwambia mwanawe. Andika katika usemi wa taarifa. Al 2

15. Eleza matumizi mawili ya mshazari. Al 2

16. Tunga sentensi yenye muundo ufuatao.
Nomino ya dhahania ,kivumishi kimilikishi, kitenzi halisi na kielezi. Al 2

17. Tambua vipashio vya lugha ukianzia na cha kiwango cha juu kisha uvitiolee mifano.

Al 4

18. Andika kwa wingi al 2

Duka lililofunguliwa asubuhi lilifungwa magharibi.

E. ISIMU JAMII AL 10

Taja na ueleze sifa za sajili ya kituo cha polisi

F. FASIHI SIMULIZI

1. Andika umuhimu wa nyiso katika jamii. Al 5

2. Eleza sifa za maigizo. Al 5

3. Tambua dhima ya mighani katika jamii. AI 5

JINA:.....NAMBARI.....DARASA:...

KISWAHILI KIDATO CHA TATU

KARATASI YA PILI

LUGHA

MUDA: 2hrs 30min

MWISHO WA MUHULA WA KWANZA 2020

MAAGIZO: Jibu Maswali yote. Andika majibu yako katika nafasi ulizoachiwa katika kijitabu hiki cha maswali.

1. Ufahamu (ala 15).

Soma taarifa ifuatayo kisha ujibu maswali yanayofuata.

Kwa kipindi cha miezi kadha iliyopita tumeshuhudia vyombo vya dola vikitia makali yake kwenye upekuzi na hata kupiga doria usiku na mchana katika jitihada za kulinda nchi.

Mpango huu ni kweli umeonekana kufanya kazi hivi kwamba hata magaidi wenyewe wameshidwa kupenya katika miji na sasa hiari kijinga kushambulia magari ya abiria, kitendo ambacho ni cha kuonyesha uwoga. Kwa hatua hito, navipa vyombo vya dola kongele. Jamabo ambalo lafaa kujulikana ni mwamba mikakati ya kulinda nchi haifai kuwa ni ya wakati mmoja tu, mbali inafaa kuwa ni zoezi la kila siku.

Magaidi nao huwa macho huku yakijua bayana kwamba wakati wa kulala kwa walinzi unapokuwepo, basi wanapata nafasi ya kututupia ‘viazi’ ukipenda grunedi.

Kama ilivyo kawaida katika mataifa mengi barani Afrika, ni bayana kwamba bado kungali na mianya mingi ambayo magaidi wa kimataifa huendelea kutumia. Dosari bado zipo. Kwa mfano, mipaka mingi ya nchi hizi huwa kama lango kuu la ugaidi wa kimataifa, kwani kuenea kwa saratani ya ufisadi halimo tu maofisini mbali pia kwenye mipaka yetu.

Kama kupata kitambulisho, pasipoti na stakabadhi zingine za kusafiri nchini bila usumbufu wowote ule bora tu anayehitaji ana hela mkononi? Kwa kuikubali hongo kuwa ufunguo wa kila kitu, Wakenya wenzangu hapo naona ni kama tumejiweka kwenye kikaango kilicho juu ya moto mkali. Hapa hakuna aliye na bahati, tajiri kwa masikini wamo kwenye mtego huu hatari.

Kwa mtindo ambao tunafuata wa kutoa ajiraa katika idara mbalimbali za ulinzi, inabidi serikali iwe na uangalifu sana hasa kwenye suala nzima la kuhakikisha stakabadhi wanazohitaji si ghushi.

Pasina kufanya hivyo hapo tena tunaweza kuwapata waadui wanaopenya na kujifanya walinzi wetu kumbe ni majasusi wa magaidi. Kila Mkenya anafaa kujihisi kulindwa. Miji, vijiji na hata vitongozi vinafaa kuwa na usalama wa kutosha, kwani kila Mkenya ni mlipa ushuru na hatufai kuona labda tabaka la juu likipendelewa huku mitaa ya mabanda ikiachiwa mbwa koko kama walinzi wao.

Suala lingine muhimu ni kuangaziwa upya usalama kwenye magari ya usafiri.

Juzi tulishuhudia mabasi mawili yakilipuwa kwenye barabara ya Thika huku tukijua fika kwamba, mpango wa walinda nchi ungalipo.

La kusikitisha ni kuona kwamba, madereva na utingo wao walikamatwa na kufunguliwa mashtaka kwa kutozuia shambulizo hilo. Je, hii ni sharia gani? Dereva ataendesha gari au atachukua jukumu la walinda usalama?

Waswahili walinena kwamba ukubwa ni jaa na kwa hivyo Rais wa taifa ndiye anayefaa kubeba mzigo mzima wa usalama wetu bila kubananga wasaa. Wengine ambao wanafaa kuwajibika ni wakuu wote wa idara mbalimbali za usalama.

Usalama wako na wangu ni muhimu, elewa bayana kwamba bila usalama watalii hawawezi kuja kututembelea. Bila ya usalama maendeleo ya taifa kamwe hayawezi kupatikana, ndiposa kila jitihada sharti zifanywe ili wote waweze kuendelea kuyafurahia matunda ya uhuru wetu.

Ada ya mja kunena muugwana ni vitendo, hatufai tu kuimba wimbo wa linda nchi' ilhali mabasi barabarani hayana usalama. Hebu na tuuone ulinzi endelevu na hapo tutawakomoa magaidi kwa yakini.

Maswali

1. Ipe taarifa hii anwani mwafaka? (alama 1)

2. Kwa nini magaidi wanashambulia magari ya abiria? (ala 1)

3. Kulingana na taarifa pamoja na maoni yako, eleza njia tatu ambazo zinatumiwa na magaidi ili kufanikisha utekelezaji wa unyama wao. (ala 3)

4. Ni njia gani ambazo magaidi hutumia kuingia katika nchi wanapoazimia kutekeleza uhalifu? (ala 3)

5. Kwa nini suala la usalama kwenye magari ya usafiri ni muhimu? (ala 1)

6. Ni njia zipi zinazoweza kutumiwa kupunguza mashambulizi ya kigaidi kulingana na mwandishi?
(ala 2)

7. Eleza athari za utovu wa usalama. (ala 2)

8. Eleza neno au mafungu ya maneno kama yalivyotumiwa katika kifungu., (ala 2)
a) Vyombo vya dola.

b) Jaa

2. UFUPISHO (ALA 15) MUHTASARI

Binadamu aliumbwa kwa njia ya kushangaza na muumba wake. Ana uwezo mkubwa wa kukumbuka matukio hata yaliyotokea miiaka na mikaka iliyopita. Pia, ana uwezo wa kukumbuka kupanga matukio yajayo.

La aajabu ni kwamba, pamoja na uwezo huu, binadamu anaendelea kukumbukwa na mabadiliko makubwa katika mazingira anamoishi. Ghafla, siku hizi, mwanadamu amejikuta katika hali ya kusahau kabisa mambo muhimu. Ameumbwa na hali ya hatihati. Amezongwa na mengi kuliko ilivyokuwa zamani.

Hali hii mpya imepelekea mwanadamu kufikiria njia ya kukidhi mahitaji yake ya kukumbuka na kufanya yote atakavyo ama atakikanapo kufanya hivyo. Ndipo alianzisha shajara.

Shajara ni daftari inayotumiwa kihifadhi habari muhimu nyumbani, ofisini, shuleni ama mwahali kwa kazi. Kunazo shajara za aina tofautitofauti. Kwanza, kunazo dafatari rasmi ambazo aghalabu ni kubwa na huwa zinasheheni habari nyingi za kimataifa. Shajara kama hizi huweza kuwa na habari kuhusu nchi mbalimbali za ulimwengu, ikiwemo bara la Afrika, nyakati za dunia, vipimo mbalimbali vya kawi, urefu, hali ya joto, umbali kutoka miji mikuu muhimu duaniani hadi mingine na hata siku kuu mbalimbali ulimwenguni. Shajara hizi hutumiwa ofisini.

Shajara za ofisini hutumiwa kurekodi tarehe za mikutano, kesi za ofisini, miadi ya maofisa na wateja na kadhalika. Kwa jumla, shughuli zote zihusuzo kampuni hurekodiwa humu. Pia, shajara za kiofisi ni muhimu kwani husaidia katika kuratibu masuala ya ofisini na kupangilia shughuli za wasimamizi ili kuokoa wakati.

Kuna shajara nyingine ambazo ni za kibinafi. Aina hizi za shajara huhifadhi mambo ya kibinafsi kama vile kurekodi tarehe za matukio muhimu, kuhifadhi tarehe za mikutano na miadi ili zikumbukwe kwa urahisi

kuhifadhi taarifa muhimu kamam iana ya damu, nambari za pasipoti na kitambulisho, nambari za simu za marafiki na jamaa, amwani, nambari za barua pepe, tarehe za kuzaliwa watoto na kadhalika. almuradi mambo yanayorekodiwa katika shajara hizi yamo katika kiwango cha kibinaFSI, kinyume na taarifa zinazohifadhiwa katika shajara rasmi.

Shajara vilevile huweza kuainishwa kutegemea utendakazi wake. Kunazo shajara za vita zinazotumiwa kurekodi matukio ya kivita. Vilevile, kunazo shajara za kisayansi ambazo hutumiwa kurekodi uvumbuzi wa kisayansi. Shuleni pia wanafunzi hutumia shajara kurekodi mambo muhimu kuhusu kazi za darasani na vitabu muhimu muhimu vya kusoma. Kwa mfano, mtihani wa KCSE unapokaribia, wanafunzi wengi hupanga ratiba ya marudio yao ili kuhakikisha kila somo limepewa uzito ufaao kulingana na tarehe ya kufanywa mtihani Fulani. Zote hizi na aina nyingine za shajara hizo.

Soma taarifa ifuatayo kisha ujibu maswali.

1. Ni nini kilimwelekeza binadamu kuanzisha shajara (maneno 30) (ala 7)
Jaribio

Jibu

2. Shajara ina manufaa yepi kwa mwanadamu? (maneno 30-45) (ala 8)
Jaribio

Jibu

3. SARUFI (ALAMA 40)

1. (a)(i) Eleza maana ya sauti? (ala 1)
- (ii) Kuna sauti ngapi katika Kiswahili? Zitaje (ala 1)
- (iii) Andika sauti mbili za ufizi. (ala 1)
- (iv) Taja sifa mbili za shadda. (al 1)
- (v) Onyesha mizizi ya vitenzi vifuatavyo. (ala 2)
- Alikuwa Watafiwa.
2. Geuza maneno yaliyopigwa mstari yawe katika hali uliyopewa
- (i) Alinitia hasira aliposema hivyo. (kitenzi) (ala 1)
- (ii) Mimi nimekuona mradi uneniona tuondoke. (nomino) (ala 1)
3. Ainisha mofimu katika kitenzi hiki. (ala 3)
- Alilolikulia

4. Amrisha vitenzi vifuatavyo. (ala 3)
- (i) Fa
 - (ii) Pa
 - (iii) La
5. Sentensi hii ina maana mbili ziandike
Huyu amekuja kutuliza. (al 2)
- (i)
 - (ii)
6. Tumia neno hili kama kielezi. (ala 1)
- Uganda
7. Bainisha virai katika sentensi hii (ala 2)
- Mtoto mwema ni aliyenipa chakula
8. Andika katika hali ya ukubwa. (ala 2)
- Mlima huu una nyoka walio na mikia mrefu.
9. Tunga sentensi moja ukitumia kirejeshi cha mbali kidogo na ngeli ya U-1. (ala 1)
10. Yakinisha sentensi hii. (ala 1)
- Leo hatuchezi mpira wa vikapu.

11. Taja majukumu ya kiimbo.

(al 4)

12. Tambulisha matumizi ya – ji- katika sentensi hii

(al 4)

Jino la jibwa hilo lilifanya lijilie chakula kingi kuliko mkimbiaji Yule.

- (i) Ji
- (ii) Ji
- (iii) Ji
- (iv) Ji

13. Andika neno lenye muundo huu :kiambishi ngeli, wakati uliopita: kirejeshi, mtendwa, mzizi, kauli, kiishio.

(ala 1)

14. Andika sentensi vile umeagizwa.

(ala 1)

- (i) Muutu aliwalea watoto kwa maki

(ala 1)

(Anza kwa watoto.....)

15. Tumia tu katika sentensi, kama

(al 3)

- i) Kielezi

- ii) Kiwakilishi

- iii) kitenzi

16. Badilisha neno lililopigwa mstari liwe kivumishi

(ala 1)

Uaminifu wa mtu humfanya aheshimiwe

17. Tumia kiwakilishi nafsi badala ya maneno yaliyopigiwa mstari. (ala 1)
Shama, tegua na koru ndio walioibuka washindi.
18. Unganisha sentensi kwa kuanzia; (Wanafunzi)wageni wamefika. Wanafunzi wamefurahi. (ala 1)

4. ISIMU JAMII (ALA 10)

- a) Eleza maana ya Isimu jamii. (al 2)
- b) Jadili dhima za lugha ya Kiswahili katika jamii (ala 2)
- c) Binadamu aweza kuwasiliana bila kutumia lugha. Eleza njia zozote tatu anazoweza kutumia (ala 6)

NAME.....ADM.NO.....CLASS.....

DATE:...../...../ 2020

MATHEMATICS

2020 FORM 3 TERM 1 OPENER EXAMS

TIME: 2½ HRS.

INSTRUCTION TO STUDENTS:

1. Write your **name**, **admission number** and **class** in the spaces provided above.
2. Write the **date** of examination in spaces provided.
3. This paper consists of **two** Sections; Section **I** and Section **II**.
4. Answer **ALL** the questions in Section **I** and only **five** questions from Section **II**.
5. All answers and working must be written on the question paper in the spaces provided below each question.
6. Show all the steps in your calculation, giving your answer at each stage in the spaces provided **below** each question.
7. Marks may be given for correct working even if the answer is wrong.
8. KNEC Mathematical tables **may be** used, except where stated otherwise.
9. Candidates should check the question paper to ascertain that all the pages are printed as indicated and that no questions are missing.
10. Candidates should answer the questions in English.

FOR EXAMINER'S USE ONLY:

SECTION I

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	TOTAL

17	18	19	20	21	22	23	24	TOTAL

SECTION II

GRAND TOTAL

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Ensure that all the pages are printed and no question(s) are missing

SECTION 1 (50 marks)

Answer all the

questions in this section in the spaces provided.

1. Evaluate

$$\frac{\frac{1}{2} \text{ of } 18 \div -3 + 2\frac{1}{2} \times \frac{3}{-5}}{\frac{1}{2} + 3\frac{3}{4} \div \frac{3}{4}}$$

(3 marks)

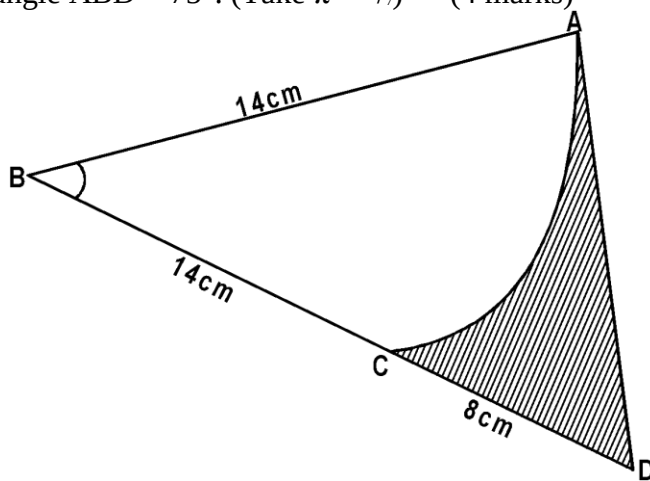
2. Solve the following without using mathematical tables or calculator.

(3 marks)

$$\frac{3600}{\sqrt[3]{27000}}$$

3. A square based brass plate is 4mm high and has a mass of 1.05kg. The density of the brass is 4.2g/cm^3 , calculate the length of the plate in centimeters. (3 marks)

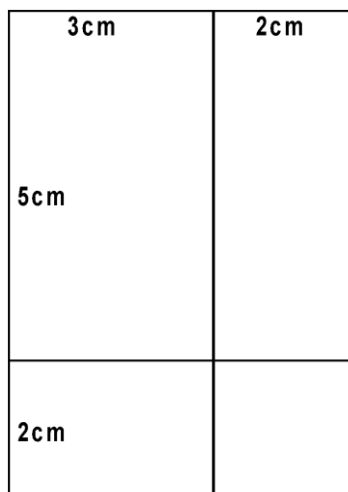
4. Calculate the area of the shaded region given that AC is an arc of a circle centre B. $AB = BC = 14\text{cm}$. $CD = 8\text{cm}$ and angle $ABD = 75^\circ$. (Take $\pi = \frac{22}{7}$) (4 marks)



5. A milk vendor bought 20 litres of milk at shs 15 per litre and added 5 litres of water to the milk. If he sold the mixture at shs 18 per litre, what was his percentage profit?

(3 marks)

6. Complete the figure below so as to make the net of a cuboid. Hence determine the surface area of the cuboid.
(4 marks)



7. Solve for x in the equation. $125^{(x+1)} + 5^{3x} = 630$ (3 marks)

8. Simplify $\frac{6x^2 - 8xy + 2y^2}{18x^2 - 2y^2}$ (3 marks)

9. A man walks directly from point A towards the foot of a tall building 240m away. After covering 180m, he observes that the angle of elevation of the top of the building is 45° . Determine the angle of elevation of the top of the building from A. (3 marks)

10. In a regular polygon, the size of an interior angle is 20° more than three times the size of its exterior angle. Find the number of sides in this polygon. (3 marks)

$$\frac{6x^3 - 8x^2y + 2xy^2}{18x^3 - 2xy^2}$$

11. A Kenyan bank buys and sells foreign currencies at the exchange rates shown below.

	Buying (Ksh)	Selling (Ksh)
1 Euro	147.56	148.00
1 Us Dollar	74.22	74.50

An American arrived in Kenya with 20,000 Euros. He converted all the Euros into Kenyan Shillings at the bank. He spent Ksh 2510200 while in Kenya and converted the remaining Kenya shillings into US Dollars at the bank. Find the amount in dollars that he received.

(3 marks)

12. Solve the following inequalities and represent the solution on a single number line.

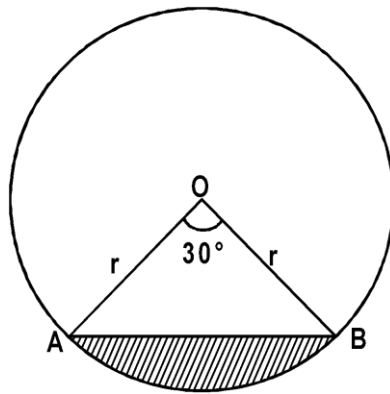
$$4 - 3x \geq -8$$

(3 marks)

$$3 - 2x < 5$$

13. A cylinder of radius 14cm contains water. A metal solid cone of base radius 7cm and height 18cm is submerged into the water. Find the change in height of the water level in the cylinder.
(4 marks)

14. The figure below show a circle centre O. Chord AB subtends 30° at the centre. If the area of the minor segment is 5.25cm^2 , find the radius of the circle. (3 marks)



15. A point $(-5, 4)$ is mapped onto $(-1, -1)$ by a translation. Find the image of $(-4, 5)$ under the same translation.
(2 marks)

16. Evaluate using Logarithms

(4mks)

$$\frac{(0.0021 + 0.0035)^{\frac{1}{2}}}{1.38 \times 27.42}$$

SECTION II (50 marks)

Answer only FIVE questions in this section in that spaces provided.

17. A matatu left Kibwezi at 7.00 am and travelled towards Nairobi at an average speed of 60km/hr. A car left Nairobi at 9.00am and travelled towards Kibwezi at an average speed of 80km/hr. The distance between the two towns is 324km.

Find:

a) the time each vehicle arrive at their destination.

i) matatu

(2 marks)

ii) car

(2 marks)

b) i) the distance the matatu covered before the car started to move from Nairobi to Kibwezi.

(1 mark)

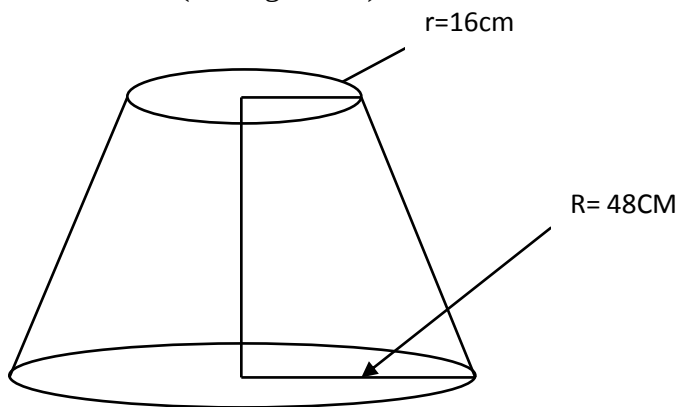
ii) the time the two vehicles met on the way.

(3 marks)

c) How far the car was from Kibwezi when they met.

(2 marks)

18. The figure below represents a frustrum of a solid cone of base radius 48 cm and top radius 16cm. The height of the frustrum is 21cm (Taking $\pi = \frac{22}{7}$) calculate:



a) The height of the solid cone.

(2 marks)

b) The volume of the solid frustrum.

(3 marks)

c) The total surface area of the frustrum.

(5 marks)

19. Towns B is 102km on bearing of 112° from town A. Town C is 94km on a bearing of 062° from town B. Town D is 073° from town A and 336° from town C.

a) Using a scale of 1cm rep. 20km, draw a diagram to show the positions of towns A, B, C and D.
(4 marks)

b) Using the diagrams in (a) above determine

i) Bearing of town B from town D.

(1 mark)

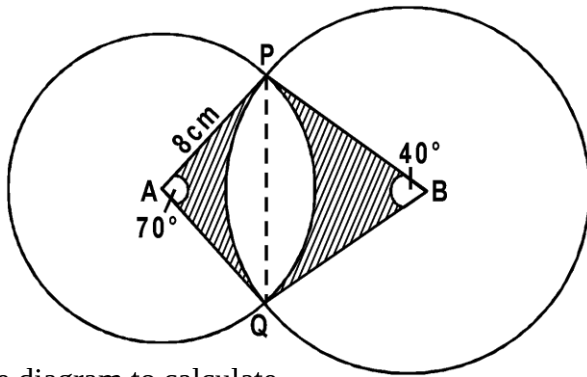
ii) Bearing of town A from town C.

(1 mark)

c) The distance AC and BD in km.

(4 marks)

20. The figure below shows two circles, centre A and B which intersect at point P and Q. Angle PAQ = 70° and angle PBQ = 40° and PA = AQ = 8cm



Use the diagram to calculate

a) PQ correct to 2 decimal places.

(2 marks)

b) PB correct to 2 decimal places.

(2 marks)

c) Area of the minor segment of the circle whose centre is A.

(2 marks)

d) Area of the shaded region.

(4 marks)

21. A salesman sold 300 bags of maize to a retailer at Kshs .2000 each .He was given a commission of 3%.The salesman allowed a discount of 0.2% on the maize sold. This discount was deducted from his commission.

(a) Calculate

(i) The discount allowed (2mks)

(ii) The net commission the salesman got (3mks)

(b)The retailer sold all the bags of maize at Ksh, 2400 each and paid Ksh 12000 for transport. Calculate the profit made by the retailer (3mks)

(c) In addition, a value added tax (V.A.T) of 16 % was charged on the profit made by the retailer. Calculate the amount of tax collected. (2mks)

22. A line L passes through points $(-2, 3)$ and $(-1, 6)$ and is perpendicular to a line P at $(-1, 6)$.
a) Find the equation of L. (2 mks)

b) Find the equation of P in the form $ax + by = c$, where a, b and c are constant. (2 mks)

c) Given that another line Q is parallel to L and passes through point $(1, 2)$ find the x and y intercepts of Q. (3 mks)

d) Find the point of intersection of lines P and Q.

(3 mks)

23. (a) Using a ruler and a pair of compasses only, construct triangle ABC and ABD on either side of line AB below, such that; $\angle DAB = \angle DBA = \angle ABC = \angle BAC = 60^\circ$ (3mks)



(b)(i) Name the quadrilateral ADBC

(1mk)

(ii) Construct a circle touching all the sides of the quadrilateral

(3mks)

(d) Calculate, correct to one decimal place, the area of the region enclosed by the quadrilateral but outside the circle.
(3mks)

24. a) On the Cartesian plane below, draw the quadrilateral PQRS with vertices.
 $P(4, 6)$, $Q(6, 3)$, $R(4, 4)$ and $S(2, 3)$ (1 mark)



- b) Draw $P'Q'R'S'$ the image of PQRS under the transformation defined by the translation vector
Write down the coordinates of $P'Q'R'S'$. (2 marks)
- c) $P''Q''R''S''$ is the image $P'Q'R'S'$ when reflected in the line $y = 1$. On the same plane, draw $P''Q''R''S''$ (2 marks)
- d) Draw $P'''Q'''R'''S'''$ the image $P''Q''R''S''$ when reflected in the line $y - x = 0$ (2 marks)
- e) Find by construction, the centre of the rotation the maps $P'''Q'''R'''S'''$ onto PQRS and hence determine the coordinates of the centre of the rotation and the angle of the rotation. (3 marks)

END TERM 1-2020.

FORM 3.

MATHEMATICS PAPER 2.

TIME: $2\frac{1}{2}$ HOURS.

NAME

ADM NUMBER

DATE.....

-Write your name, admission number and exam date on the spaces above.

-This paper consists of 13 printed pages. Check the question paper to ascertain that all the pages are printed as indicated and that no question is missing.

-Answer ALL the questions in the spaces provided.

For examiner's use only

Section I

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Score																

Section II

Question	17	18	19	20	21	22	23	24
Score								

Total score

SECTION I (50 MARKS)

Attempt ALL the questions in this section in the spaces provided.

1. Simplify (3marks)

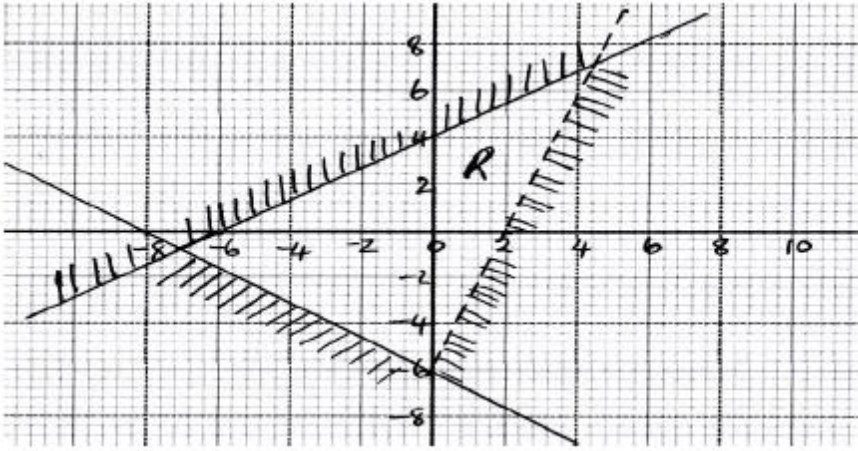
$$\frac{2\sqrt{2}+1}{2\sqrt{2}-1}$$

2. Given that $\cos(x - 20)^\circ = \sin(2x + 32)^\circ$ and x is an acute angle, find $\tan(x - 4)^\circ$ (3marks)

3. Find a quadratic equation where roots are $1.5 + \sqrt{2}$ and $1.5 - \sqrt{2}$, expressing it in the form $ax^2 + bx + c = 0$, where a , b and c are integers (3marks)

4. The length and breadth of a rectangular flower garden were measured and found to be 4.1m and 2.2m respectively. Find the percentage error in its area (3marks)

5. Determine the inequalities that define the unshaded region in the diagram below. (3mks)



6. Solve the following simultaneous equations (3mks)

$$x + y = 3$$

$$6y - xy = 70$$

7. Given that $\log 5 = 0.6990$ and $\log 7 = 0.8451$, find $\log 24.5$ (3mks)

8. Simplify completely. (3mks)

$$\frac{2x^2 + x - 3}{4x^2 - 9}$$

9. A 890kg culvert is made of a hollow cylindrical material with outer radius of 76cm and an inner radius of 62cm. It crosses a road of width 3m. Determine the density of the material used in its construction in Kg/m^3 correct to one decimal place. (3marks)

10. In a mixed school there are 900 students; out of these 600 are girls.

(a) Find the ratio of boys to girls. (2marks)

(b) What is the percentage of boys in this school? (1mark)

11. Lines K whose equation is $2y - 6 = 4x$ is perpendicular to another line Q. Find the equation of line Q if it passes through point. $(-2, 7)$. (3 marks)

12. The marked price of a car in a dealers shop was Ksh.450 000. Kawira bought the car at 7% discount. The dealer still made a profit of 13%. Calculate the amount of money the dealer had paid for the car to the nearest shillings. (3marks)

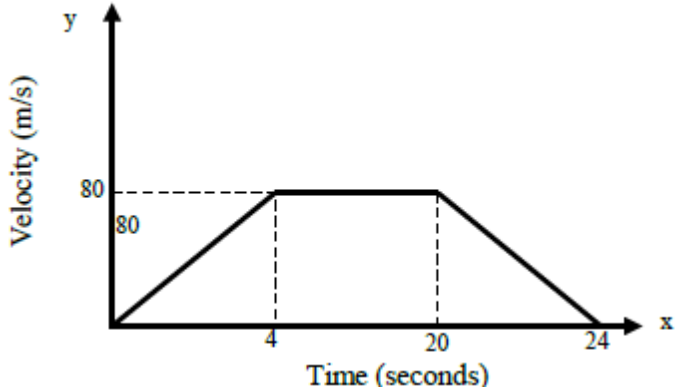
13. Given that $8 \leq y \leq 12$ and $1 \leq x \leq 6$, find the maximum possible value of; (3marks)

$$\frac{y+x}{y-x}$$

14. The length of a rectangle is $(x+1)$ cm. its width is 3cm shorter than its length. Given that the area of the rectangle is 22cm, find its length using completing the square method. (4marks)

15. A car uses 1 litre of petrol for every 8 kilometres. The car was to travel 480 kilometres and had 15 litre of petrol at the beginning of the journey. Each litre of petrol cost sh. 112.00. How much did it cost for the extra petrol added? (3mks)

16. The figure below is a velocity – time graph for a car. (not drawn to scale).



a).Find the total distance traveled by the car? (2 Marks)

(b) Calculate the deceleration of the car. (2 Marks)

SECTION II (50 MARKS)

Answer ANY FIVE questions in this section.

17) In a triangle ABC, BC=8cm, AC=12cm and angle ABC=120°

a) Calculate the length of AB, correct to one decimal place (4marks)

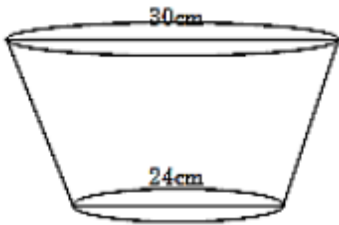
b) If BC is the base of the triangle, calculate correct to one decimal place

i. The perpendicular height of the triangle (2marks)

ii. The area of the triangle (2marks)

iii. The size of angle ACB (2marks)

18. The diagram below shows a frustum which represents a bucket with an open end diameter of 36cm and a bottom diameter of 24cm. The bucket is 30cm deep as shown and it is used to fill an empty cylindrical tank of diameter 1.4m and height 1.2m.



(a) leaving π in your answer , calculate

(i) The capacity of the bucket in litres. (6mks)

(ii) The capacity of tank in litre. (2mks)

(b) Determine the number of buckets that must be drawn in order to fill the tank. (2mks)

19. The table below shows the age groups and number of people who are HIV/ AIDSpositive, in a certain sub county in Kenya.

Age group	10 – 19	20 – 29	30 – 39	40 – 49	50 – 59	60 – 69	70 – 79
No. of people	12	15	16	25	18	10	4

- a).State the modal age group. (1mark)
- b) Calculate the mean age of the people who are HIV/AIDS Positive. (3marks)
- c) Calculate the median of the age group. (3marks)
- d) Draw on the grid provided a histogram to represent the above information. (3marks)

(The grid to be provided by the examiner)

20. The diagram below shows a solid made of a hemisphere and a cylinder. The radius of both the cylinder and the hemisphere is 3cm. The length of the cylinder is 12cm.



a).i) Calculate the volume of the solid. (3marks)

ii) The solid fits in a box in the shape of a cuboid 15 cm by 6cm by 6cm. Calculate the volume of the box not occupied by the solid correct to four significant figures. (2marks)

b) i) Calculate the total surface area of the solid correct to four significant figures. (3marks)

ii) The surface of the solid is to be painted. One millilitre of paint covers an area of 8cm^2 . The cost of paint is Ksh 900 per litre. Calculate the cost of the paint required. (2marks)]

21. Members of a group decides to raise k£ 100 towards a charity. Five of them were unable to contribute. Each of the rest had therefore to pay k£ 1 more, in order to raise the same amount.

a) If the original number of member was x, writes downs:

i) An expression of how much each was originally to contribute. (1mark)

ii) Two distinct expressions of how much each contributed after the five pulled out.(2marks)

b) Calculate the value of x. (3marks)

c) Solve the equation (4marks)

$$\frac{y+3}{24} = \frac{1}{y-2}$$

22. Arc of a circle of radius 40cm subtends an angle of 126° at the centre of the circle. Using $\pi = \frac{22}{7}$;

(a) Calculate:

(i) The length of the arc. (2 marks)

ii).The area of the sector. (2 marks)

(b) The sector is folded to form a cone.

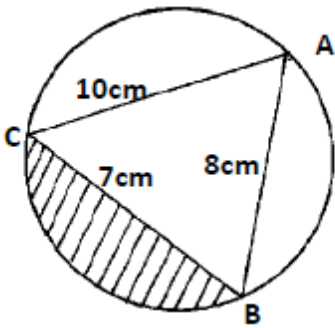
Calculate:

i).The radius of the base of the cone. (2 marks)

ii).The height of the cone. Correct to 2 decimal places (2 marks)

iii).The capacity of the cone in litres correct to 2 decimal places. (2 marks)

23. The figure below shows a triangle ABC inscribed in a circle. AC = 10cm, BC = 7cm and AB = 10cm.



a).Find the size of angle BAC. (3mks)

b).Find the radius of the circle. (2mks)

c).Hence calculate the area of the shaded region. (5mks)

24. A tailor bought a number of suits at a cost of sh.57, 600 from ken-suit wholesalers. Had he bought the same number of suits from Umoja wholesaler it would cost him sh.480 less per suit. This would have enabled him to buy 4 extra suits for the same amount of money.

(a) Find the number of suits the tailor bought. (6 mks)

(b) The tailor later sold each suit for shs. 720 more than he had paid for it. Determine the percentage profit he made. (4 mks)

NAME..... INDEX NO.....CLASS.....

PHYSICS FORM THREE**TIME 2 HOURS****INSTRUCTION TO CANDIDATES**

- a) Write your name and index number in the spaces provided above
- b) This paper consist of TWO sections A & B
- c) Answer ALL the questions in section A and B in the spaces provided.
- d) This paper consists of **9** printed pages.
- e) ALL working **MUST** be clearly shown.
- f) Mathematical tables and silent electronic calculators may be used

Take Acceleration due to gravity $g=10\text{m/s}$

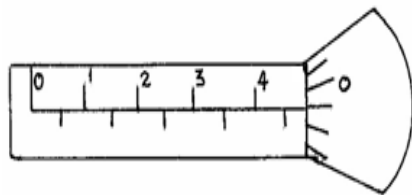
FOR EXAMINERS USE ONLY

SECTION	QUESTION	MAXIMUM SCORE	CANDIDATE'S SCORE
A	1-12	25	
B	13	11	
	14	4	
	15	7	
	16	10	
	17	12	
	18	5	
	19	6	

	TOTAL SCORE	80	
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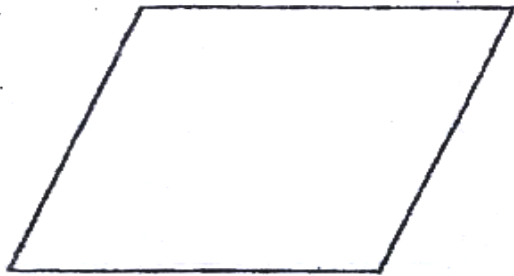
SECTION A (25 MARKS)

1. The diagram below shows a micrometer screw gauge used by a student to measure the thickness of a wire. If it has a zero error of $+0.06\text{mm}$, what is the actual thickness of the wire?
(2mks)



2. Give a reason why a thick glass bottle cracks when boiling water is suddenly poured inside it.
(1mk)
3. Other than temperature state **one** other factor that affects the surface tension of water.
(1mk)
4. State how heat losses by convection and radiation are minimized in a thermos flask.
(2mks)

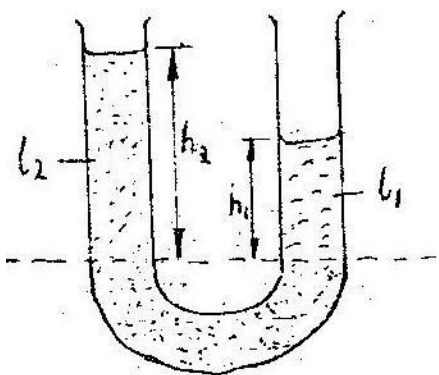
5. Figure below shows a uniform cardboard in the shape of parallelogram



Locate the centre of gravity of the cardboard.

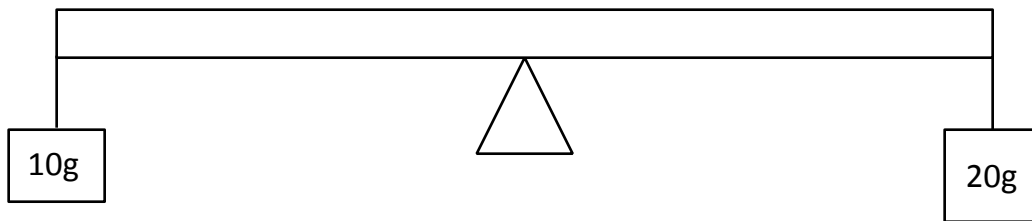
(1mk)

6. Figure below shows a U tube containing two liquids L_1 and L_2 of densities 1.8 gcm^{-3} and 0.8 gcm^{-3} respectively in equilibrium. Given that $h_2 = 8 \text{ cm}$ determine the value of h_1 . (3mks)



7. An oil drop of volume 0.2cm^3 spreads on a water surface to form a patch of diameter 30cm . determine the diameter of an oil molecule.
(3 Marks)

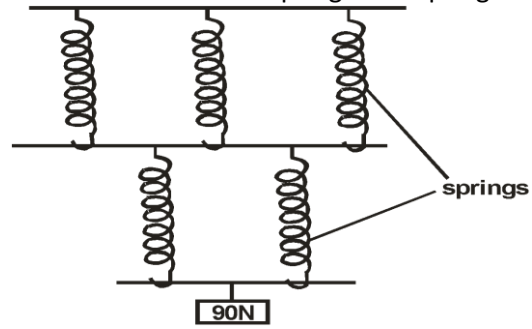
8. A uniform metre rule of negligible mass is balanced by suspending a 10g mass and a 20g mass on its ends as shown **below**.



Determine the position of the pivot.

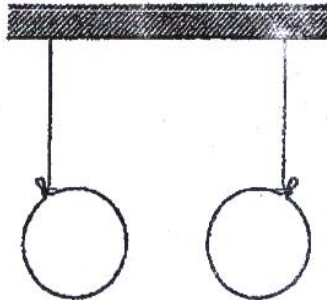
(3 marks)

9. The spiral springs shown in figure below are identical. Each spring has a spring constant $k = 300\text{N/m}$



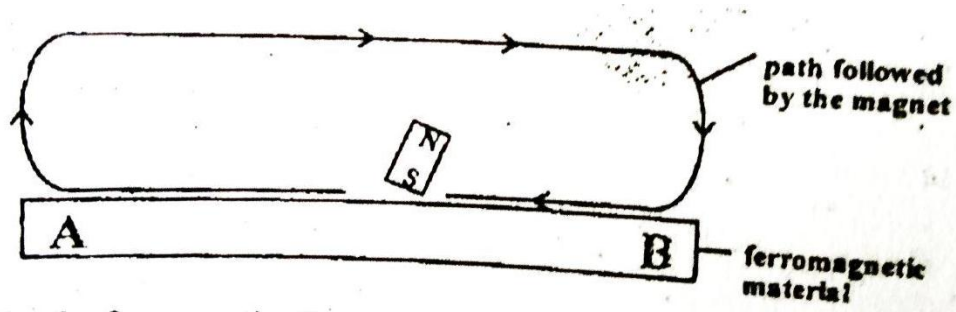
Determine the total extension of the system (take the weight of the cross bars to be negligible) (3 mks)

10. Figure below shows two inflated balloons hanging vertically on light threads.



When a stream of air is blown in the space between the balloons, they are observed to move towards each other. Explain this observations (2mks)

11. The figure below shows a method of magnetization.



i. Name the method of magnetization. (1 mk)

ii. Name the polarity acquired by end B. (1 mk)

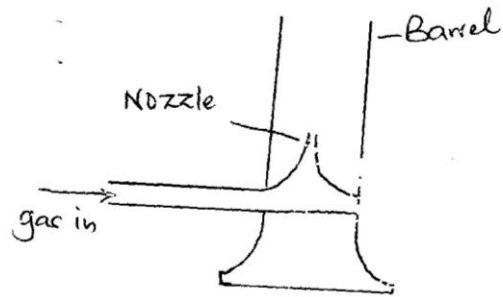
12. A car manufacturer wishes to increase the stability of a car he is designing. State two ways he can achieve this. (2 mks)

SECTION B (55 MARKS)

13. a) State Bernoulli's principle.

(1mk)

b) The diagram below shows a Bunsen burner



i. Explain how the burner works

(3mks)

ii. A pipe of radius 6cm is connected to another pipe of radius 9cm. If water flows through the wider pipe at a velocity of 2m/s, calculate;

I. Volume of water leaving the wider pipe per second ($\pi=3.142$)

(3mks)

II. Velocity of water in the small pipe

(2mks)

c) Explain why it's dangerous to stand close to a busy highway

(2mks)

14. (a) State the basic law of charges.

(1 mk)

(b) State **one** uses of an electroscope.

(1 mks)

(c) A negatively charged sphere X is brought up to an identical uncharged sphere Y. the spheres do not touch.



Sphere Y is earthed by touching it with a wire and then detached. Sphere X moved away from sphere Y.

i) What is the purpose of the insulating holder?

(1 mk)

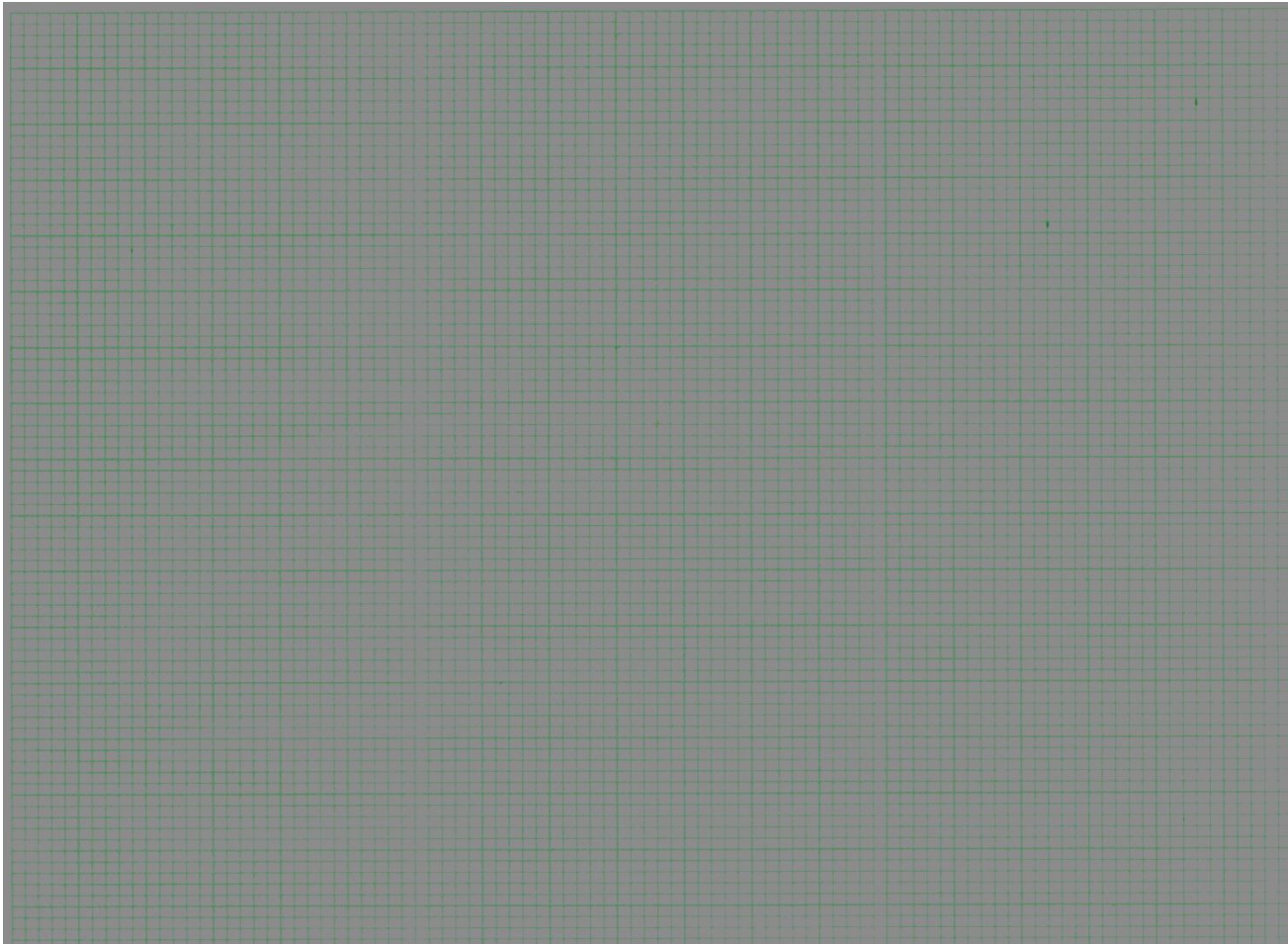
ii) State the final charge on sphere Y.

(1 mk)

15. (a) State the difference between the principal focus of a convex mirror and that of the concave mirror. (1 mks)

(b) An object of height 2cm is placed 60cm in front of a concave mirror of focal length 20cm.

i) Locate by scale drawing, the image formed. What is the image distance? (4 mks)



ii) State one characteristics of the image.

(1 mk)

iii) Determine the magnification of the image.

(1 mk)

16. a) The height of a mercury barometer at a particular place is 70cm. Given that the density of mercury is 13600kg/m^3 , determine;

i. The atmospheric pressure at the place in Pascal's. (3marks)

ii. The height of a water barometer at the same place (density of water $=1\text{ g/cm}^3$) (3marks)

iii. Give a reason why mercury is preferred as a barometric liquid. (1mk)

b) Calculate the minimum pressure a block of dimensions $3\text{cm} \times 10\text{cm} \times 15\text{cm}$ and mass 12kg could exert on a horizontal surface. (3mks)

17. a) State Hooke's law.

(1mk)

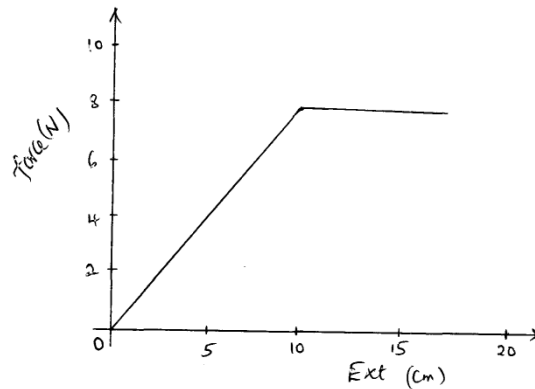
b) An object of weight 20N attached at the end of a spring causes an extension of 0.5cm on the spring

i. Determine the spring constant

(3mks)

ii. Determine the weight of an object that would cause an extension of 0.86m when attached at the end of the same spring.
(2mks)

c) One end of a piece of rubber band was fixed on a rigid support and the other end pulled with a varying force. The graph below shows the relationship between force and extension for the rubber band.



Using the graph determine

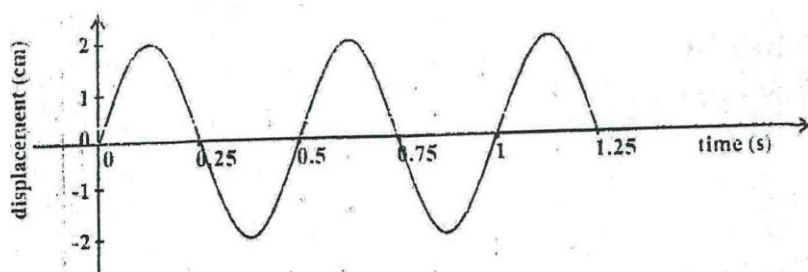
i) The stretching force at elastic limit

(1mks)

ii) The elasticity constant of the rubber band (3mks)

iii) Assuming two identical rubber bands were used instead of one and attached in series what would be the elasticity constant? (2mks)

18. The figure below shows how the displacement of a wave varies with time.



Determine the;

a) Amplitude (1 mk)

b) Frequency of the wave.

(2 mks)

c) Wavelength given that the velocity of the wave is 60cm/s.

(2 mks)

19. a) Distinguish between primary and secondary cells. (1mks)

b)i) Define polarization with reference to a simple cell. (1mk)

ii) State how polarization can be overcome in simple cells. (1mk)

c) A charge of 90 coulombs passes through a point in a circuit every one minute. Calculate the current through the point. (3marks)

END TERM 1-2020.

FORM 3

PHYSICS PAPER 1

TIME: 2 HOURS.

NAME

ADM NUMBER

DATE.....

-Write your name, admission number and exam date on the spaces above.

-This paper consists of 8 printed pages. Check the question paper to ascertain that all the pages are printed as indicated and that no question is missing.

For examiner's use only

QUESTION	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
SCORE																	

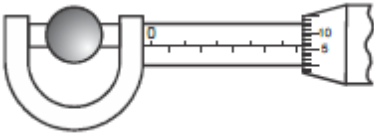
TOTAL SCORE

SECTION A: (25 MARKS)

Answer all the questions in this section.

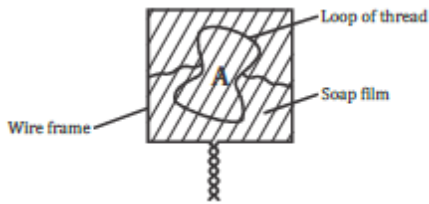
1. A stopwatch started 0.36s after the start button was pressed. The time recorded using a stopwatch for an athlete running from point A to B was 12.86s. Determine the actual time taken by the athlete. (2mks)

2. Fig 1.0 shows a spherical ball held between the anvil and the spindle of a micrometer screw gauge.



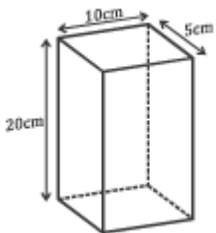
Determine the radius of the spherical ball. Give your answer in SI units. (2 marks)

3. Figure 2.0 shows a loop of cotton thread tied onto a wire frame. The figure is dipped into a soap solution and withdrawn.



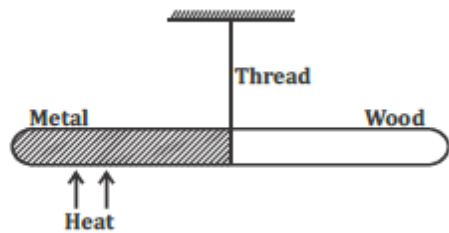
Illustrate and explain what happens to the shape of the loop of thread when part A is broken by touching it with a hot needle. (2 marks)

4. Figure 3.0 shows a brick of mass 8.0kg standing upright on the ground as shown.



What is the pressure it exerts on the ground ? ($g = 10\text{Nkg}^{-1}$) (3 marks)

5. The figure 4.0 shows a rod made of wood on one end and metal on the other end suspended freely with a piece of thread so that it is in equilibrium.



The side made of metal is heated with a bunsen burner flame. State with a reason, the side to which the rod is likely to tilt. (2mks)

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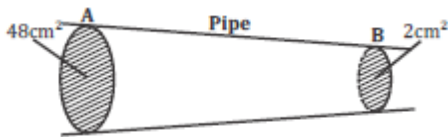
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6. State the relationship between Physics and Mathematics. (1mark)

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7. The diagram in fig 5.0 shows a section of a pipe with different cross-sectional area.



If water flows with a velocity of 10m/s in section A, what would be the velocity of water in section B ? (3 marks)

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8. Determine the spring constant of a spiral spring whose length changes from 10cm to 18cm when a load of 10.0N is suspended from its lower end. (3 marks)

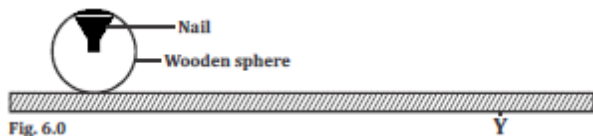
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9. Figure 6.0 below shows a wooden sphere with a nail hammered into it as shown.



The sphere is rolled on a horizontal ground and comes the rest after sometime at point Y. Draw the sphere after it comes to rest at point Y. (1 mark)

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10. When the temperature of an enclosed gas at constant pressure is raised, the volume of the gas increases. Explain how the molecules of the gas causes the increase in volume. (2 marks)

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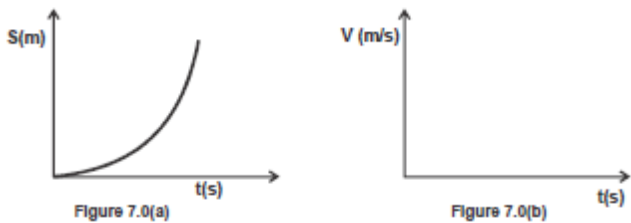
11. It is easier to stop a saloon car than a bus when both are moving with the same velocity. Explain. (2 mks)

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12. Figure 7.0 (a) shows a displacement-time graph. Sketch a velocity-time graph on fig. 7.0(b) (1mark)



13. State the law of conservation of energy.(1mark)

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SECTION B(55 MARKS).

Answer ALL the questions in the spaces provided.

14. a) State the principle of conservation of linear momentum. (1mk)

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b) Distinguish between elastic and inelastic collision. (1mk)

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c) A striker kicks a ball of mass 200 g initially at rest with a force of 78N. Given that the foot was in contact with the ball for 0.30s, determine the take – off velocity of the ball. (3mks)

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d) A high jumper usually lands on a thick soft mattress. Explain how the mattress helps in reducing the force of impact. (2mks)

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e) A ball is thrown horizontally from the top of a vertical tower of height 75m and strikes the ground at a point of 80m from the bottom of the tower. Determine the:

i) Time taken by the ball to hit the ground (acceleration due to gravity= 10m/s^2) (3mks).

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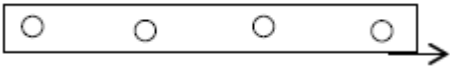
ii) Initial horizontal velocity of the ball. (2mks)

15. (a). (i) Define the term velocity and state its SI unit. (2 marks)

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(ii) The figure below show a tape obtained from a ticker timer operated at a frequency of 100 Hz.



Determine the time taken to move from one dot to the successive dot. (2 mks)

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(b) A student from Favour high school threw a stone horizontally at a velocity of 25 m/s from a top of a tree 45 m high. Calculate;

(i) The time taken by the stone to hit the ground.(take $g = 10 \text{ m/s}^2$) (3 marks)

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(ii) The distance covered by the stone horizontally from the base of the tree. (3 marks)

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(c) A car started from rest and accelerated uniformly at 2 m/s^2 for 5 seconds. It then travelled for 3 seconds with attained velocity before accelerating again at 2.5 m/s^2 for 2 seconds. The car was brought to rest within 2 seconds.

(i) Sketch a velocity - time graph for the car's motion. (3 marks)

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c) A box of mass 20 kg is opposed by a frictional force whose co-efficient is 0.8 . Calculate,

i) Frictional force on the box. (3 marks)

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ii) State two methods of minimizing frictional force. (2 mks)

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17) a) Use the table below to answer the questions that follow;

Mass m (g)	0.0	50	100	150	200	250	300
Force F (N)	0.0	0.5					
Length (cm)	34.2	35.8	37.8	38.9	40.5	42.0	43.6
Extension e (cm)							

i)Complete the table by filling in the missing values. (2mks)

(ii) Plot a graph of force (x-axis) against extension. (5mks)(The grid to be provided by the examiner)

(iii) From the graph, determine the spring constant. (2mks)

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(iv) Why will the graph not be linear if a large mass is hung? (1mks)

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b) A spring with a spring constant of 25 N/cm extends by 5 cm when a certain force acts on it. Determine the work done by the force. (2mks)

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28(a) State two structure of the skin essential for its homeostatic function. (2mks)

b) Name two kidney diseases. (2mks)

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